MS-7057

Version 100

INTEL (R) Brookdale-G/GL/GE/GV Chipset

Willamette/Northwood 478pin mPGA-B Processor Schematics

CPU: Willamette/Northwood mPGA-478B Processor

System Brookdale-G / GE / PE Chipset: INTEL GMCH + ICH4

On Board Chipset:

BIOS -- FWH

LPC Super I/O -- W83627HF-AW

Clock Generator -- CY28349

AC'97 Codec -- RealTek ALC650/655

Onboard Lan Chipset-- RealTek RTL8101L

Expansion Slots:

AGP2.0 SLOT * 1

PCI2.2 SLOT * 3

Platform: Micro ATX

MODEL Config.	ORCAD Config.	Function	Option	ERP Number
MS7057-0A	cfg6714GV-LAN	GV+LAN+Com2	A	601-7057-A10
MS7057-0A	cfg6714GE-LAN	GE+LAN+AGP+Com2	B	601-7057-A20

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AGP 4X SLOT (1.5V)	15
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AC'97 Codec and Audio Connector	18
USB Connector	19
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VRM 9 NIKO N2101	21
Realtek RTL8101L LAN	22
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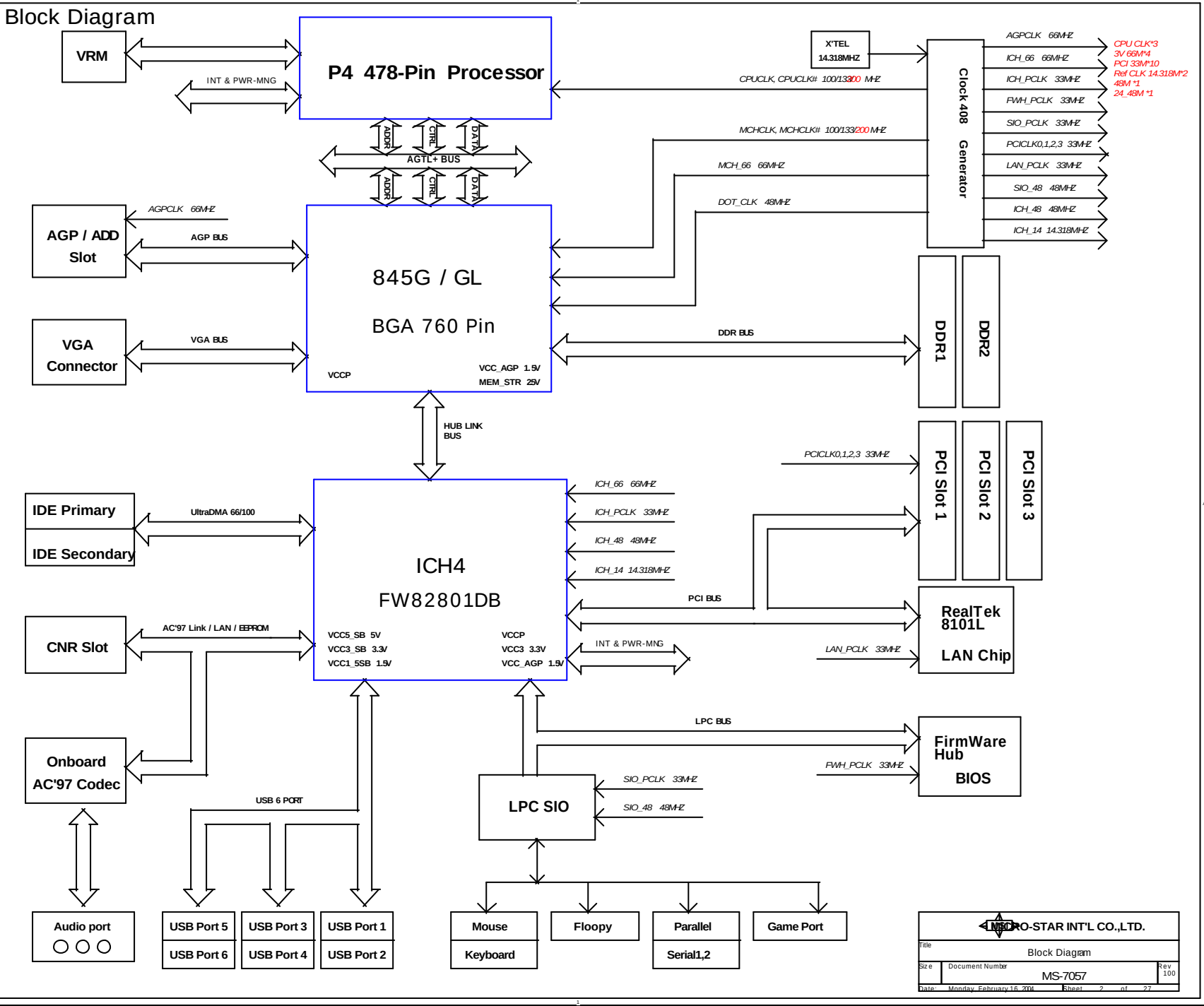
MICRO-STAR

Title: COVER SHEET

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Block Diagram

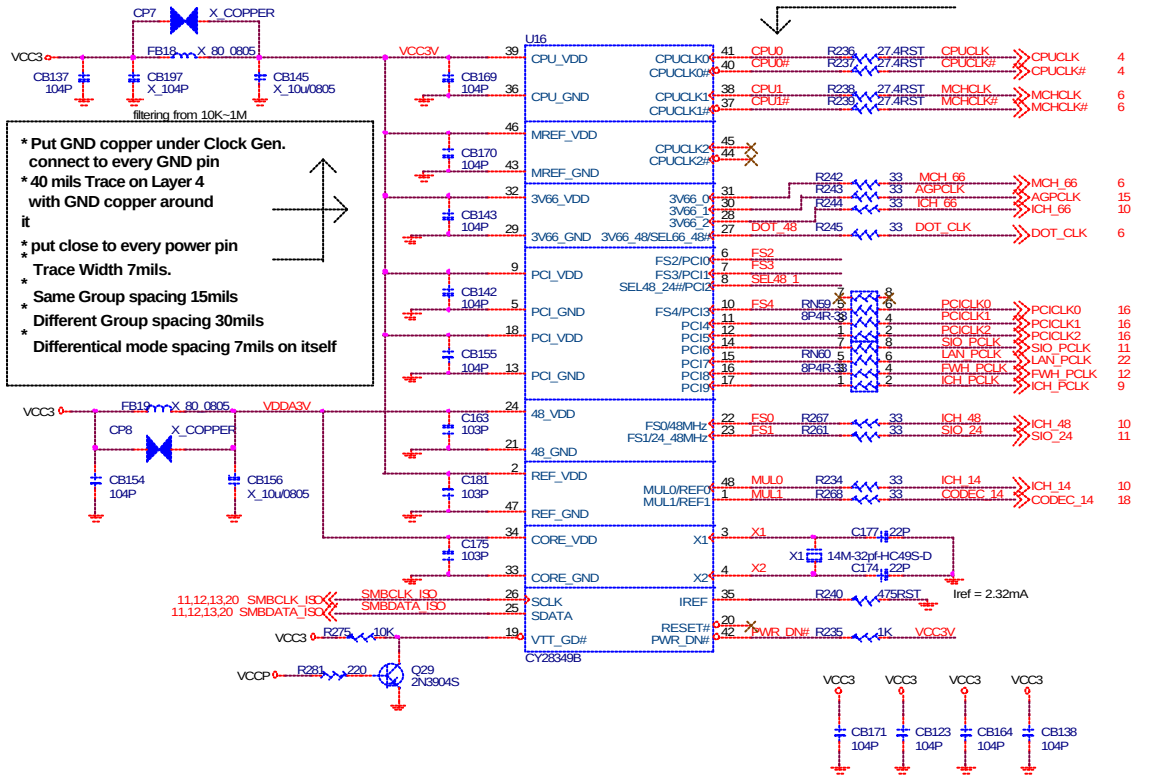


MICRO-STAR INT'L CO.,LTD.

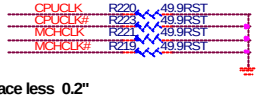
Title	Block Diagram	
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CLOCK GENERATOR BLOCK

*Trace < 0.5"



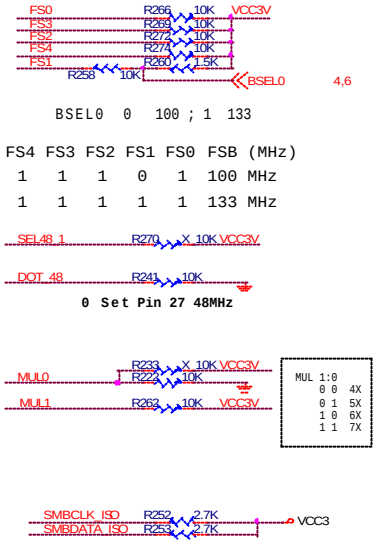
Shut Source Termination Resistors



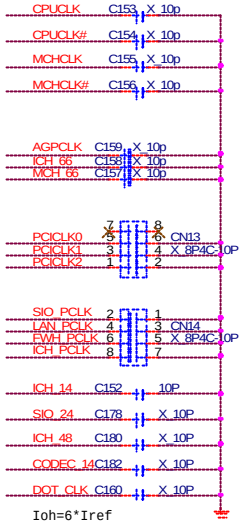
Trace less 0.2"

49.9ohm for 50ohm M/B impedance

CLOCK STRAPPING RESISTORS



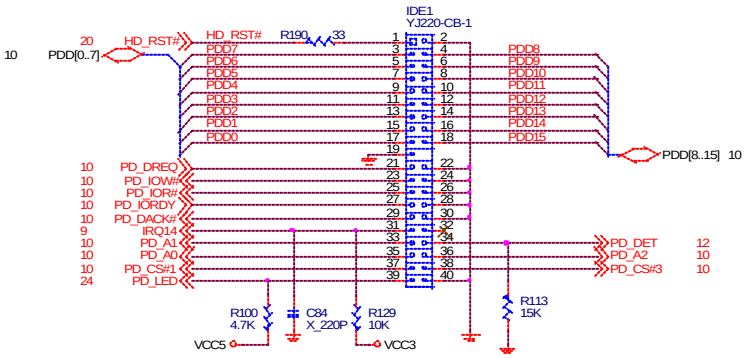
Pull-Down Capacitors



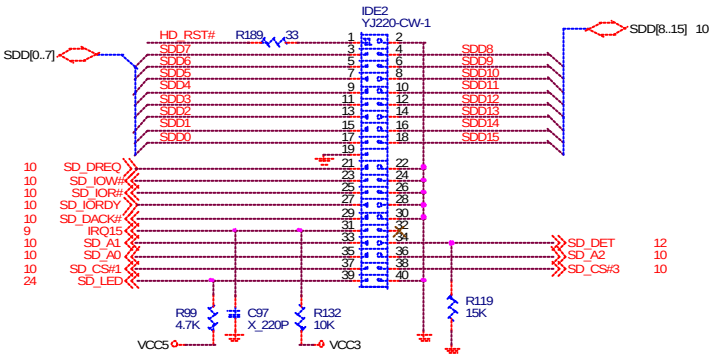
used only for EMI issue

Trace less 0.2"

PRIMARY IDE BLOCK



SECONDARY IDE BLOCK



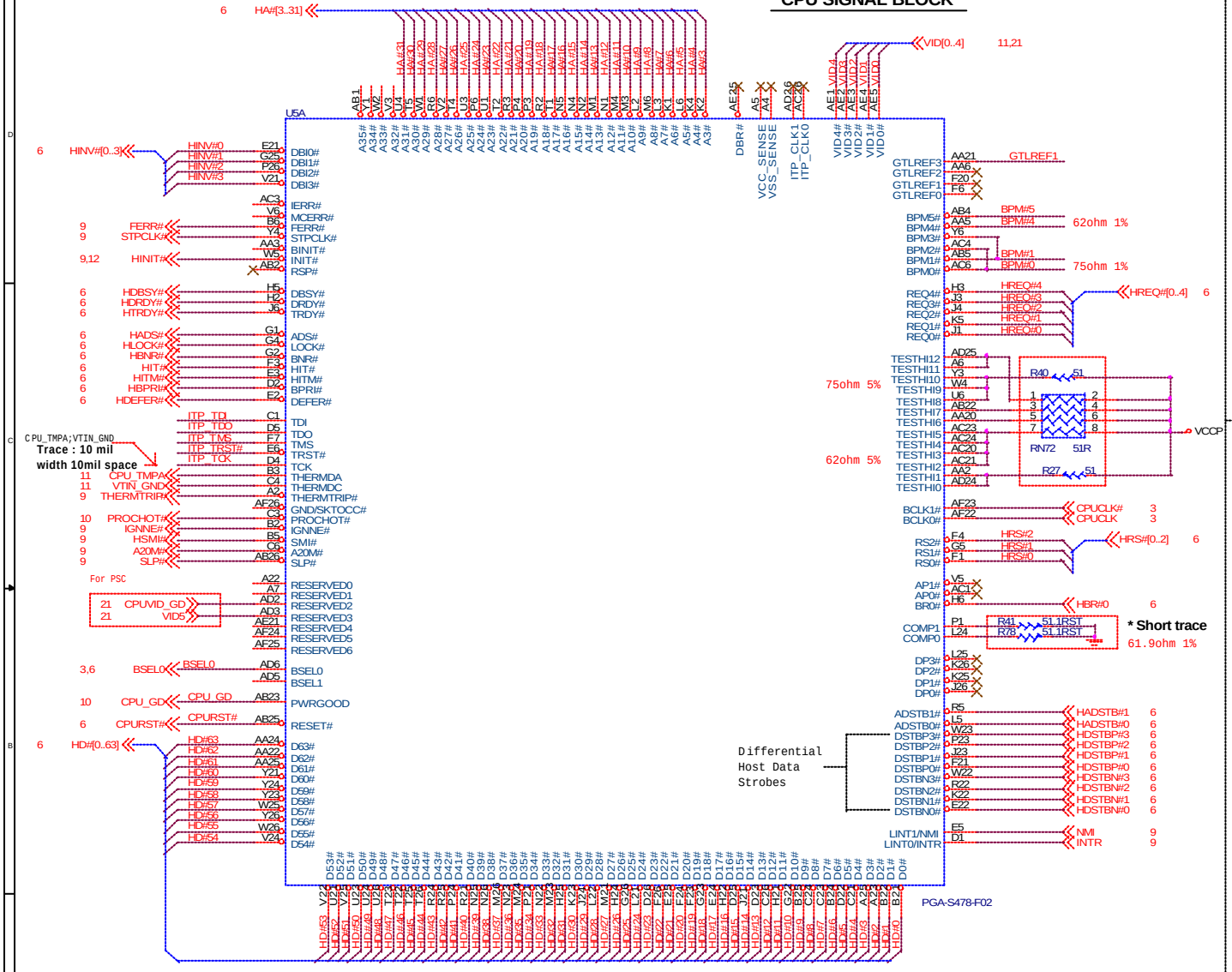
ATA100 IDE CONNECTORS

* Trace Width : 5mils
* Trace Spacing : 7mils
* Length(longest)-Length(shortest)<0.5"
* Trace Length less than5"

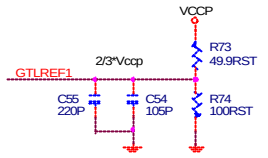
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CPU SIGNAL BLOCK



CPU GTL REFERNCE VOLTAGE BLOCK



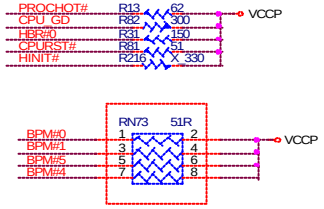
Every pin put one 220pF cap near it.
Trace Width 10mils, Space 15mils.
Keep the voltage dividers within 1.5 inches of the first GTLREF Pin

CPU ITP BLOCK



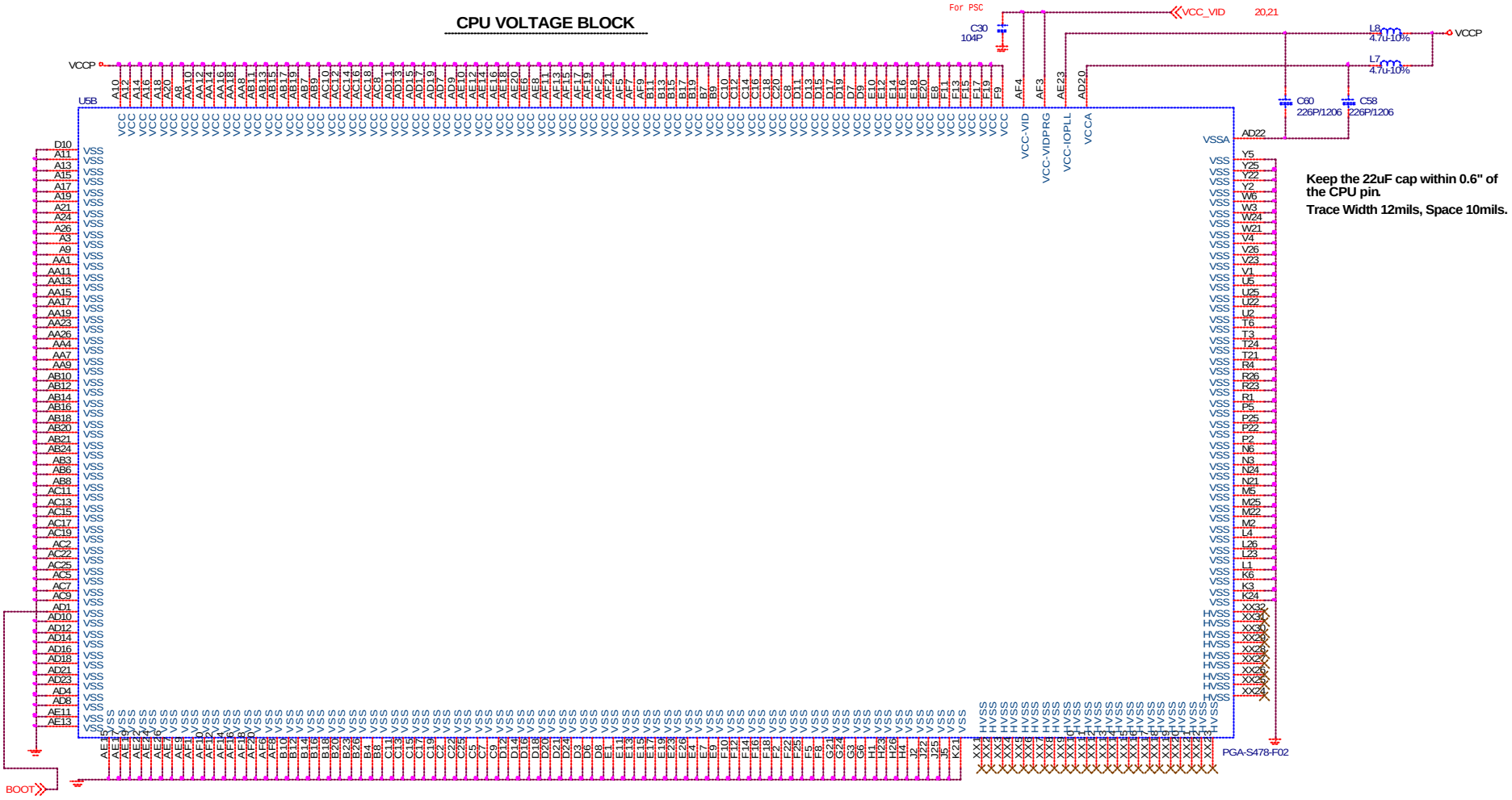
ALL COMPONENTS CLOSE TO CPU

CPU STRAPPING RESISTORS

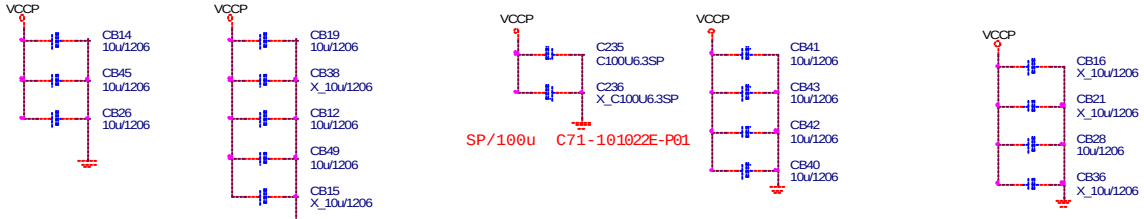


MSI	MICRO-STAR	
Title	INTEL mPG478 CPU1	
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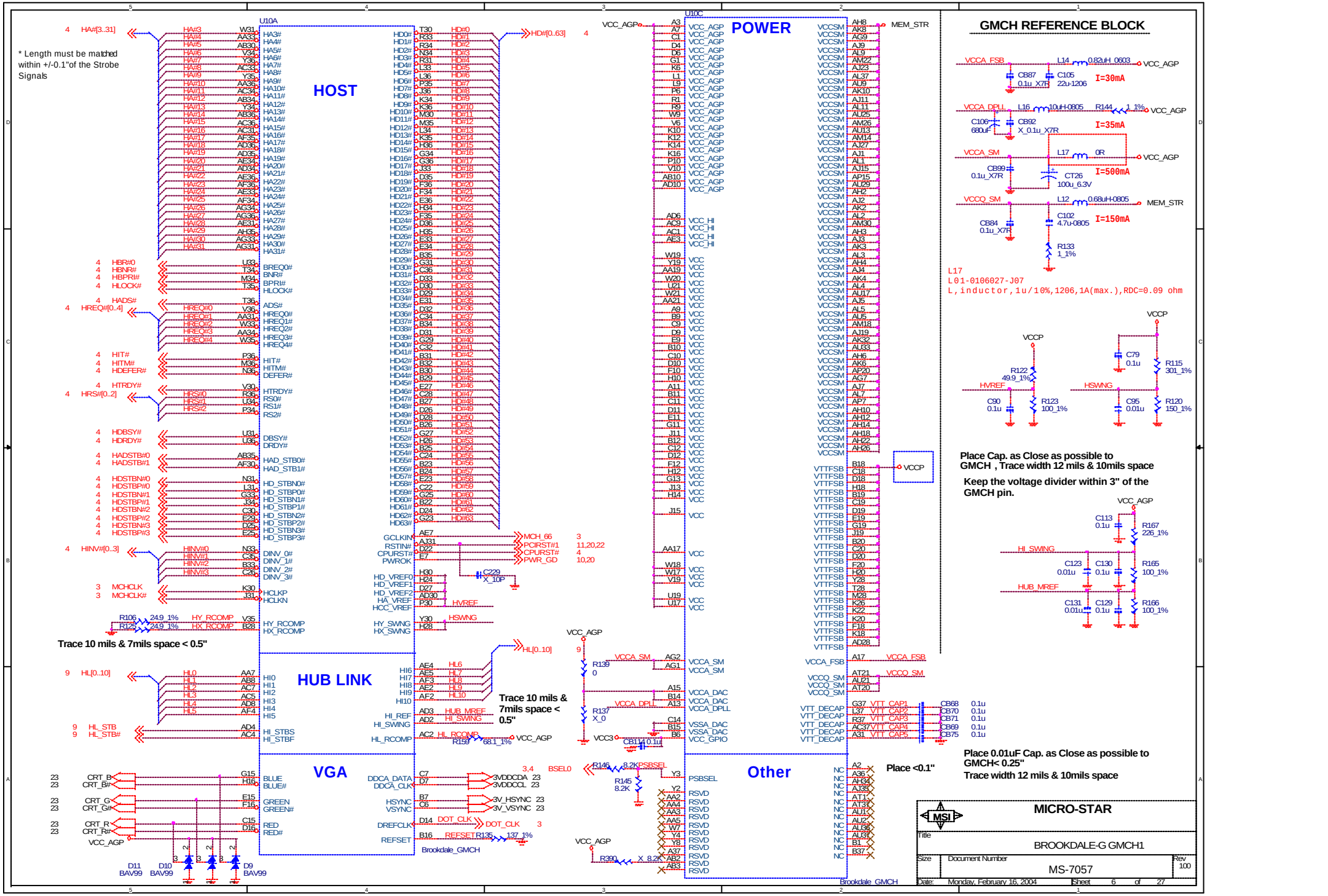
CPU VOLTAGE BLOCK

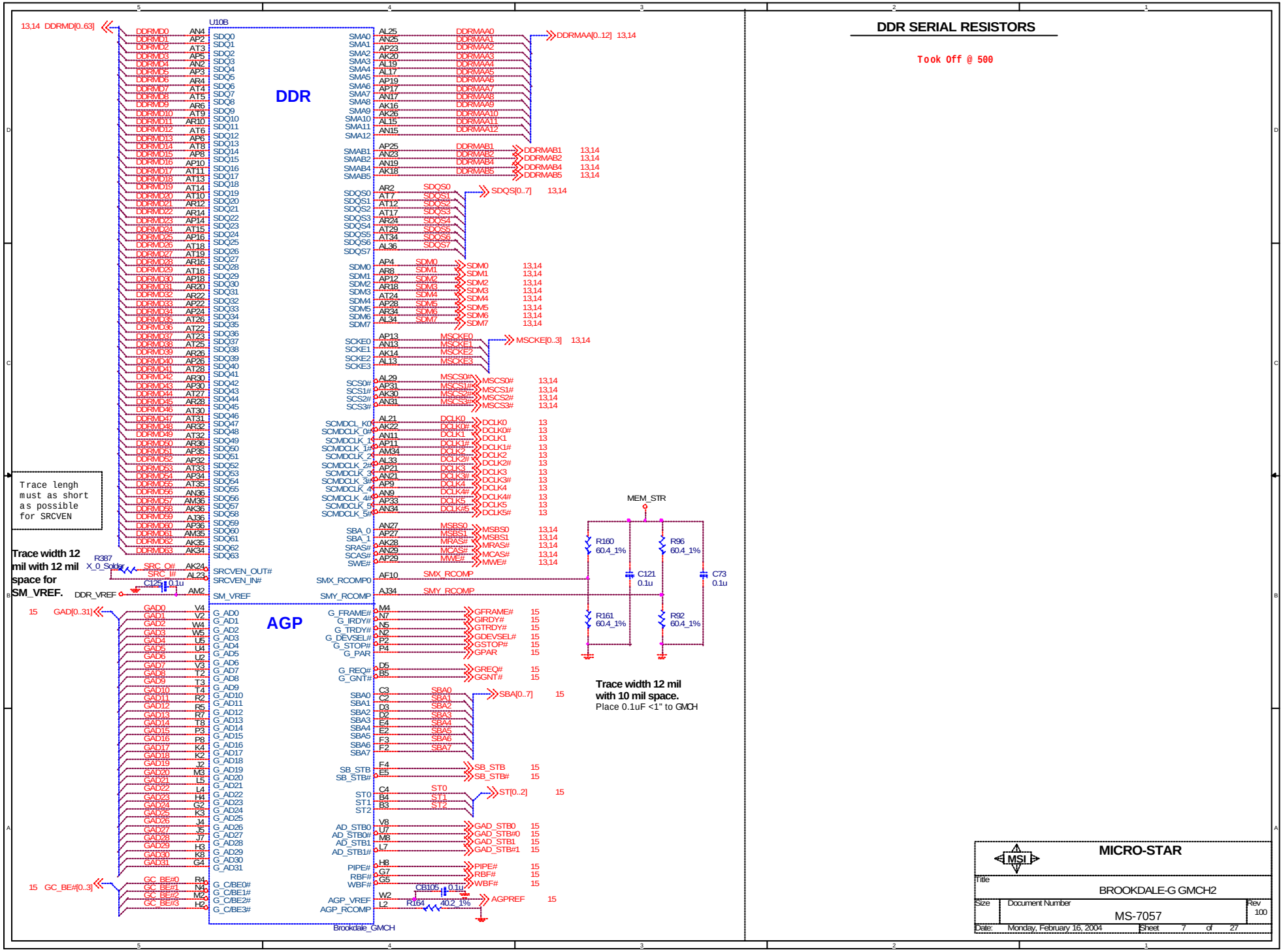


CPU DECOUPLING CAPACITORS



MICRO-STAR		
Title	INTEL mPGA478-B CPU2	
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HOST
POWER
HUB LINK
VGA
Other
GMCH REFERENCE BLOCK
* Length must be matched within +/-0.1"of the Strobe Signals
Trace 10 mils & 7mils space < 0.5"
Trace 10 mils & 7mils space < 0.5"
Place Cap. as Close as possible to GMCH , Trace width 12 mils & 10mils space
Keep the voltage divider within 3" of the GMCH pin.
Place 0.01uF Cap. as Close as possible to GMCH< 0.25"
Trace width 12 mils & 10mils space
Place <0.1"
L17
L01-0106027-J07
L,inductor,1u/10%,1206,1A(max.),RDC=0.09 ohm
MICRO-STAR
BROOKDALE-G GMCH1
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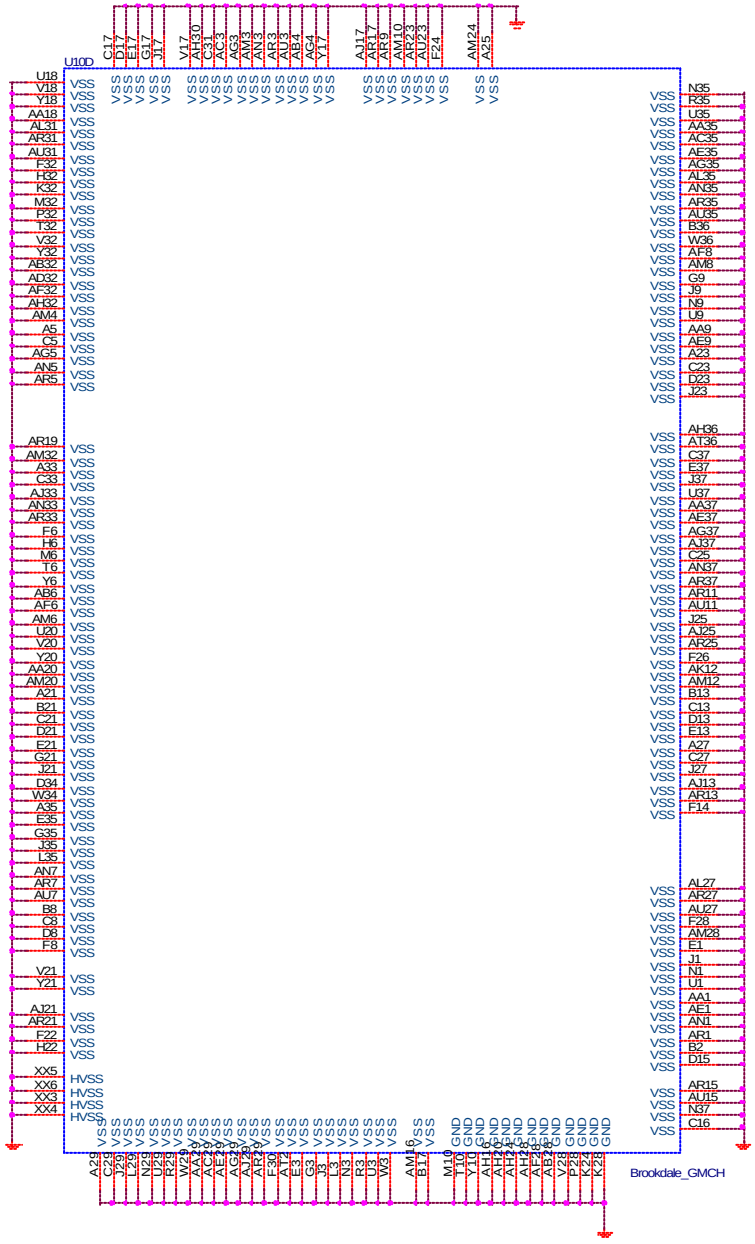


DDR SERIAL RESISTORS

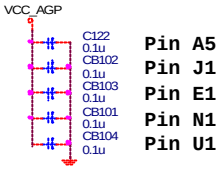
Took Off @ 500

MICRO-STAR

Title	BROOKDALE-G GMCH2	
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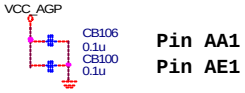


GMCH DECOUPLING CAPACITOR



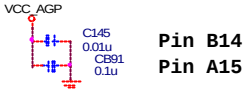
Pin A5
Pin J1
Pin E1
Pin N1
Pin U1

Place decoupling cap close to GMCH AGP Interface < 0.1"



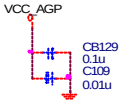
Pin AA1
Pin AE1

Place decoupling cap close to GMCH Hub-Link Interface< 0.1"

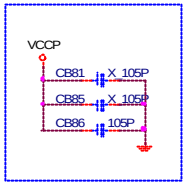


Pin B14
Pin A15

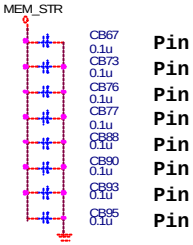
Place decoupling cap close to GMCH DAC Interface< 0.1"



Place decoupling cap close to GMCH Core Logic Interface < 0.1"

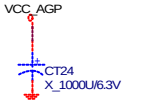


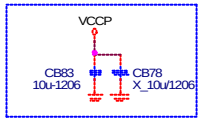
Place decoupling cap close to GMCH CPU Interface < 250mil in the Vtt corridor



Pin AL37
Pin AU33
Pin AU29
Pin AU25
Pin AU17
Pin AU13
Pin AU9
Pin AU5

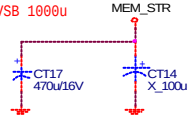
Place decoupling cap close to GMCH Memory Interface < 0.1", with 18 mil trach width





Place Bulk cap for Core Logic, AGP & Hub Link Interface

MEM_STR 470u*2 <3VSB 1000u



Place Bulk cap between GMCH & DIMM slot

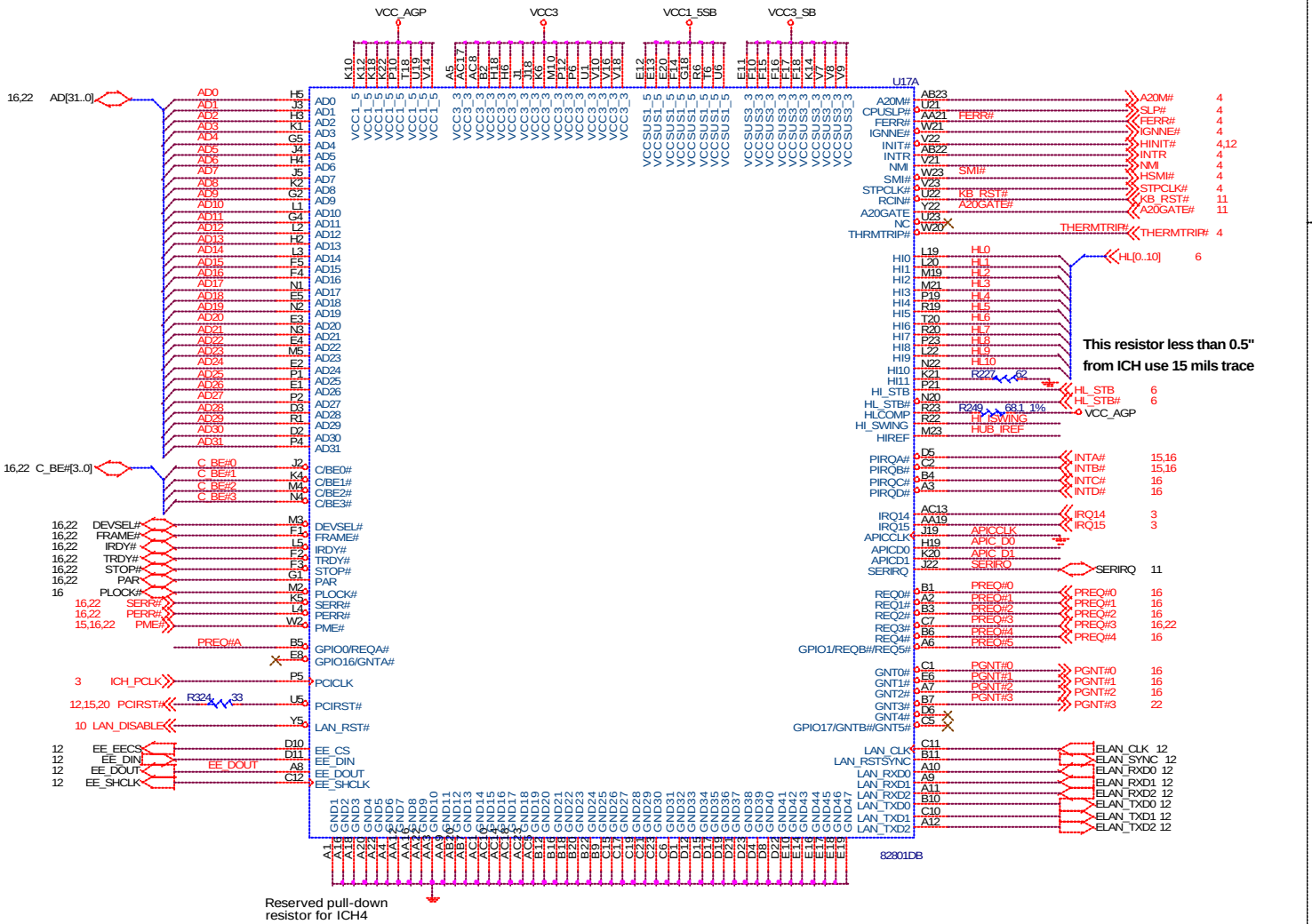
MICRO-STAR

Title: Broodale-G GMCH3

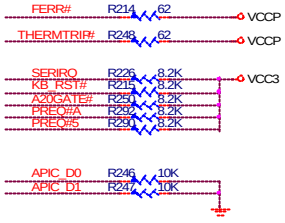
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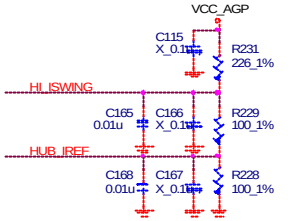
ICH4 PCI / HUB LINK / CPU / LAN / INTERRUPT SIGNALS



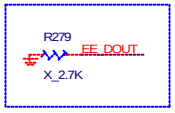
ICH4 PULL-UP/DOWN RESISTORS



ICH4 REFERENCE VOLTAGE

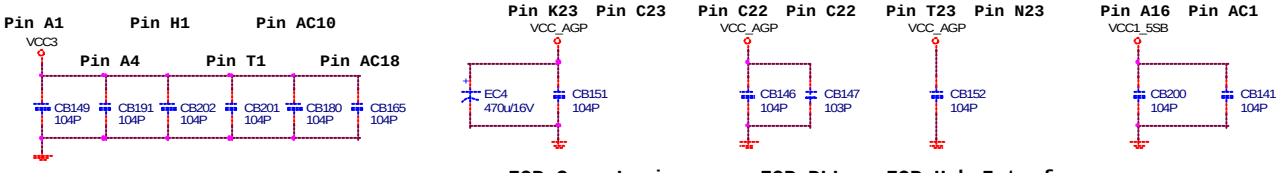


Place Cap. as Close as possible to ICH4 < 0.25"
Trace width use 12 mils and 10mils space



ICH4 DECOUPLING CAPACITORS

Place one 0.1u close to ICH4 <100 mil



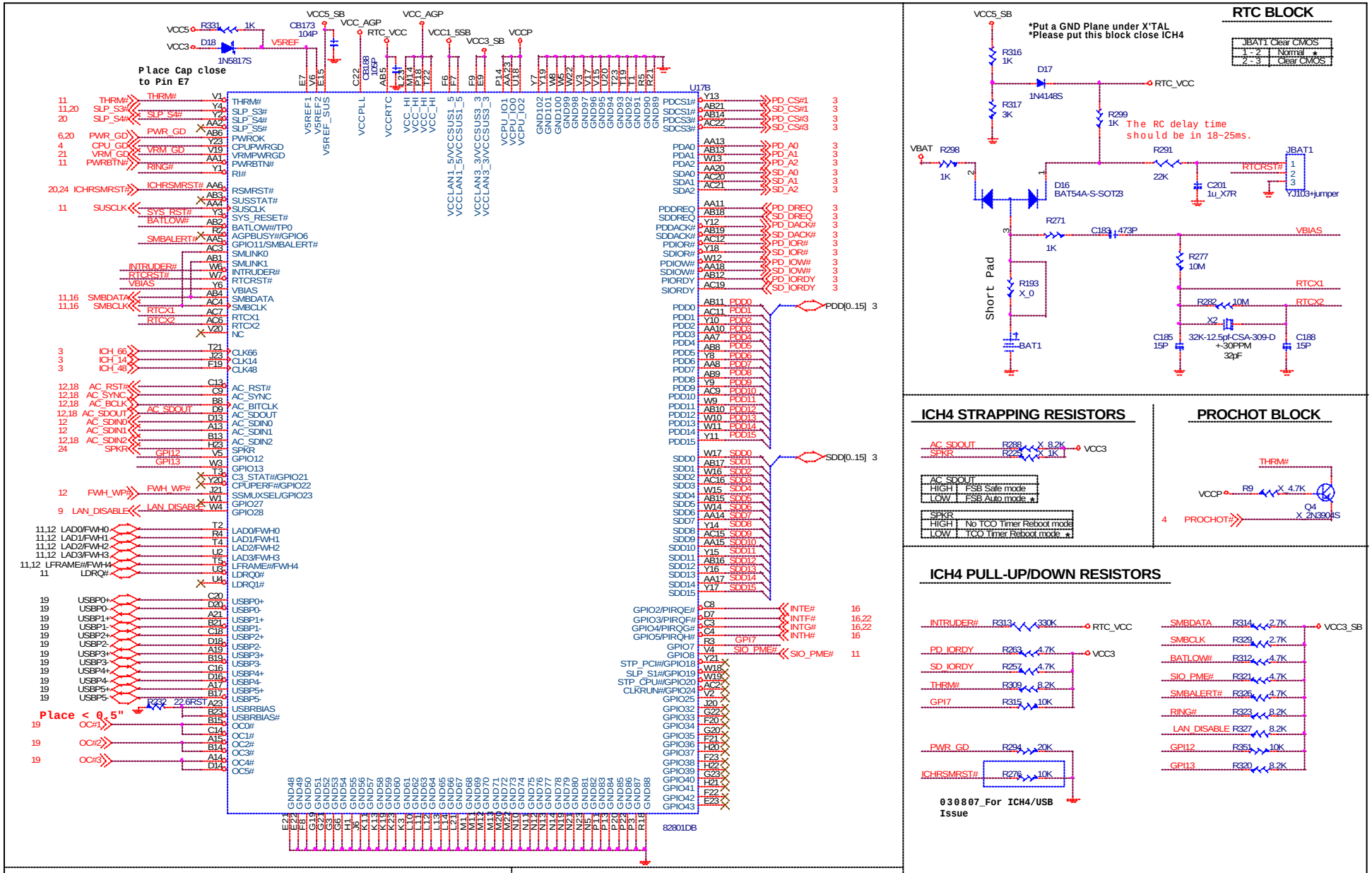
FOR Core Logic FOR PLL FOR Hub Interface

MICRO-STAR

Title: ICH4 PCI/HI/LAN

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RTC BLOCK

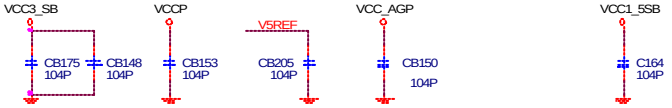
ICH4 STRAPPING RESISTORS

PROCHOT BLOCK

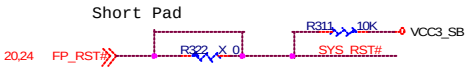
ICH4 PULL-UP/DOWN RESISTORS

ICH4 DECOUPLING CAPACITOR

Place one 0.1u close to ICH4 <100 mil



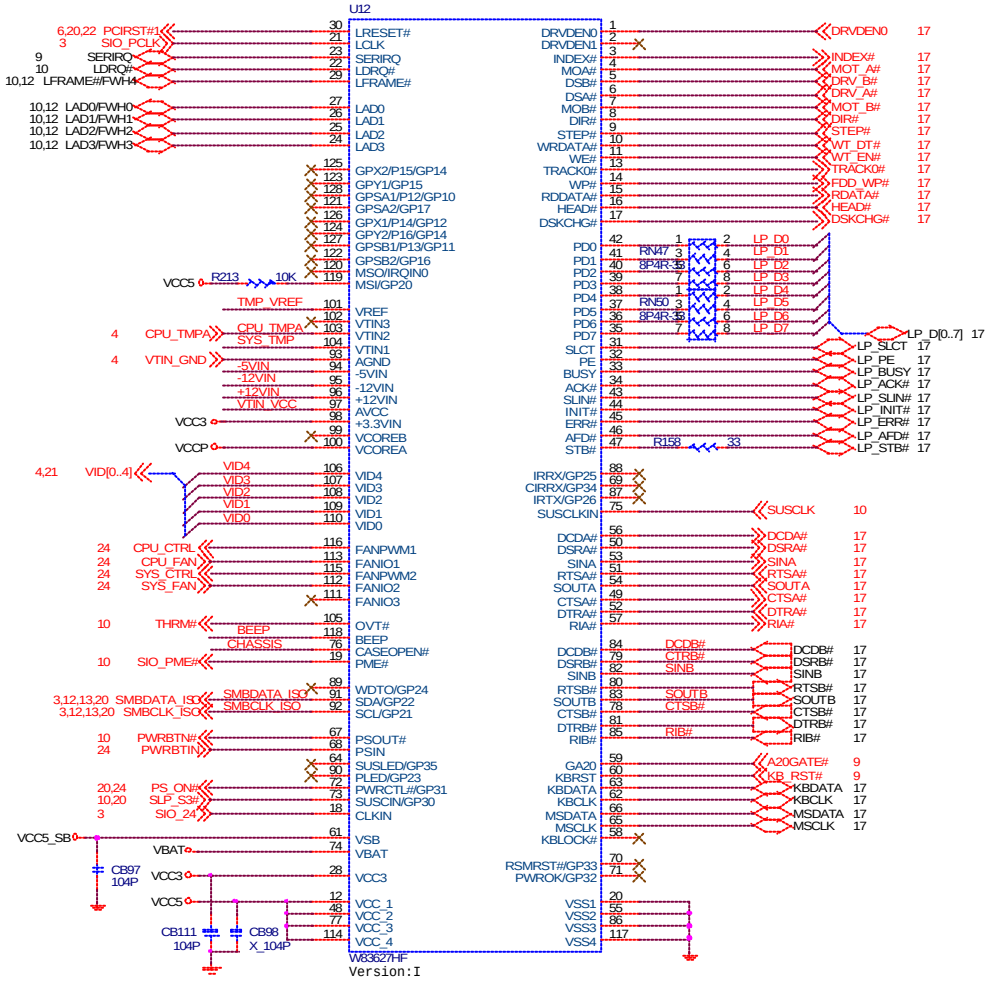
SYSTEM RESET



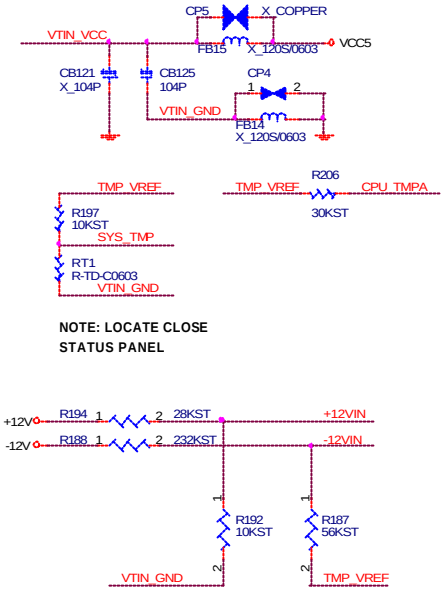
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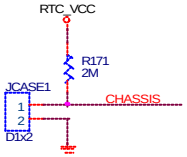
LPC SUPER I/O W83627HF



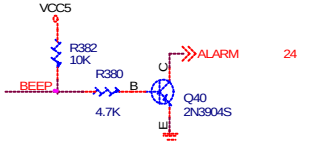
THERMAL RESISTOR BLOCK



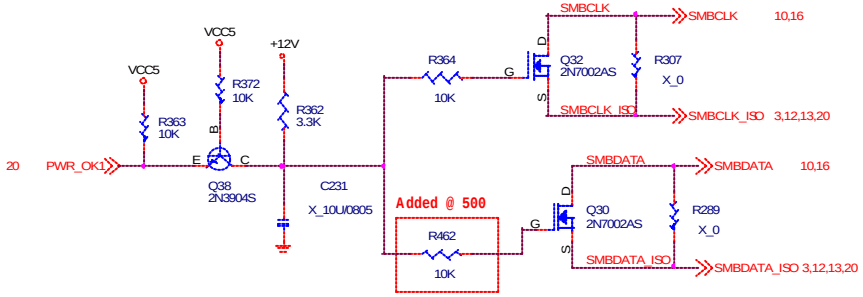
Chasiss Intrusion Header



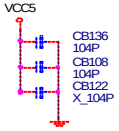
SPEAKER BLOCK



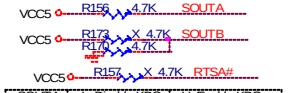
SMBus Isolation



LPC I/O DECOUPLING CAPACITORS



SUPER I/O STRAPPING RESISTOR

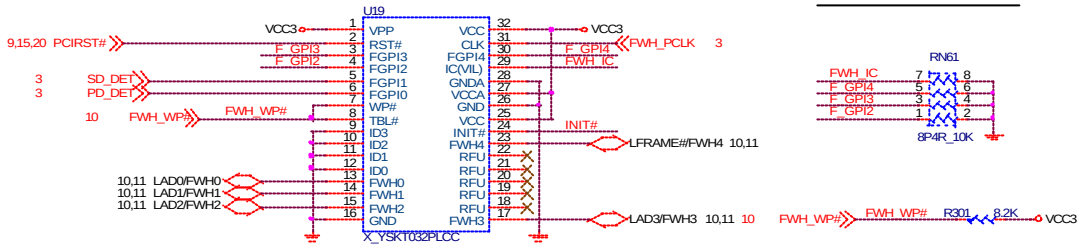


SOUTA	L: Disable KBC	H: Enable KBC
SOUTB	L: 24MHZ	H: 48MHZ
RTSA#	L: CFAD=2E	H: CFAD=4E
DTRA#	L: PNP Default	H: PNP no Default

MICRO-STAR

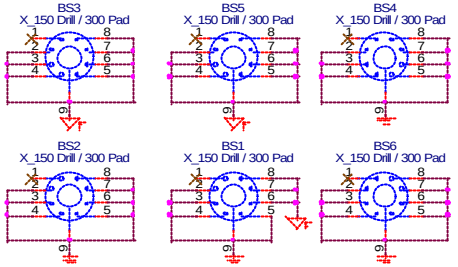
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Firware Hub (FWH)

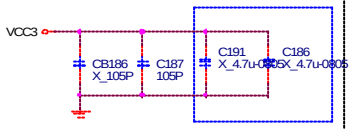


FWH RESISTORS

PCB Mounting Holes



FWH DECOUPLING CAPACITORS

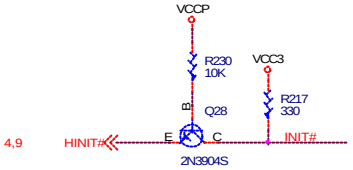


Place Cap. as Close to FWH< 350 mil

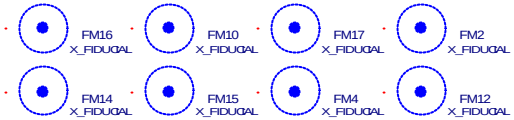
FWH write protect

	BIOS_WP BIOS Update
SHORT	Flash Write Disable
OPEN	Flash Write Enable **(Default)**

FWH INIT Signal Voltage Translation Block



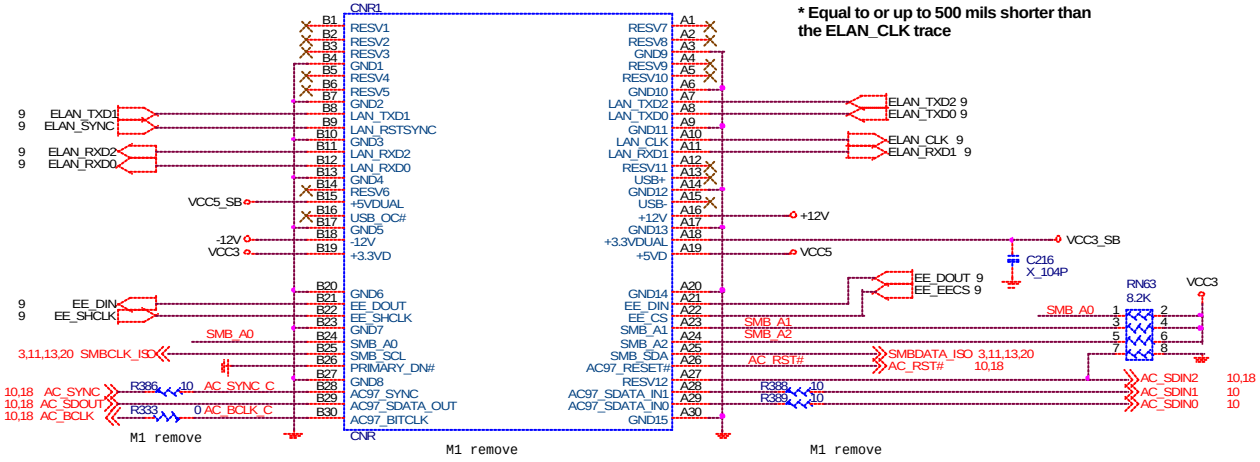
PCB Fiducials



SIMULATION TRACE



CNR RISER



*** LAN Trace width : 5 mils**

*** AC'97 Trace Spacing : 10 mils**

*** Maxium trace length < 9.5"**

*** Equal to or up to 500 mils shorter than the ELAN_CLK trace**

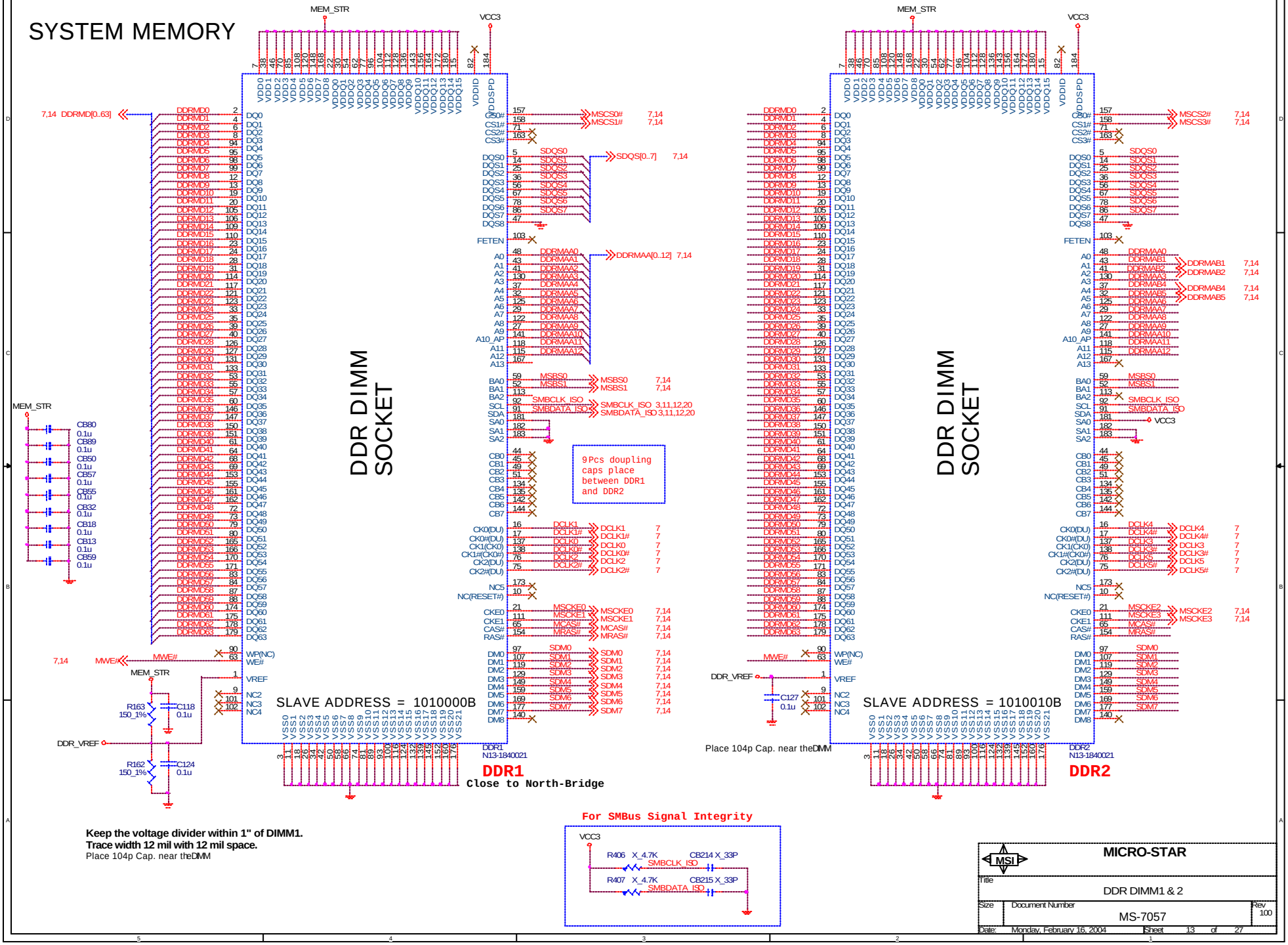
MICRO-STAR

Title: FWH

Size | Document Number: MS-7057 | Rev: 100

Date: Tuesday, February 17, 2004 | Sheet 12 of 27

SYSTEM MEMORY



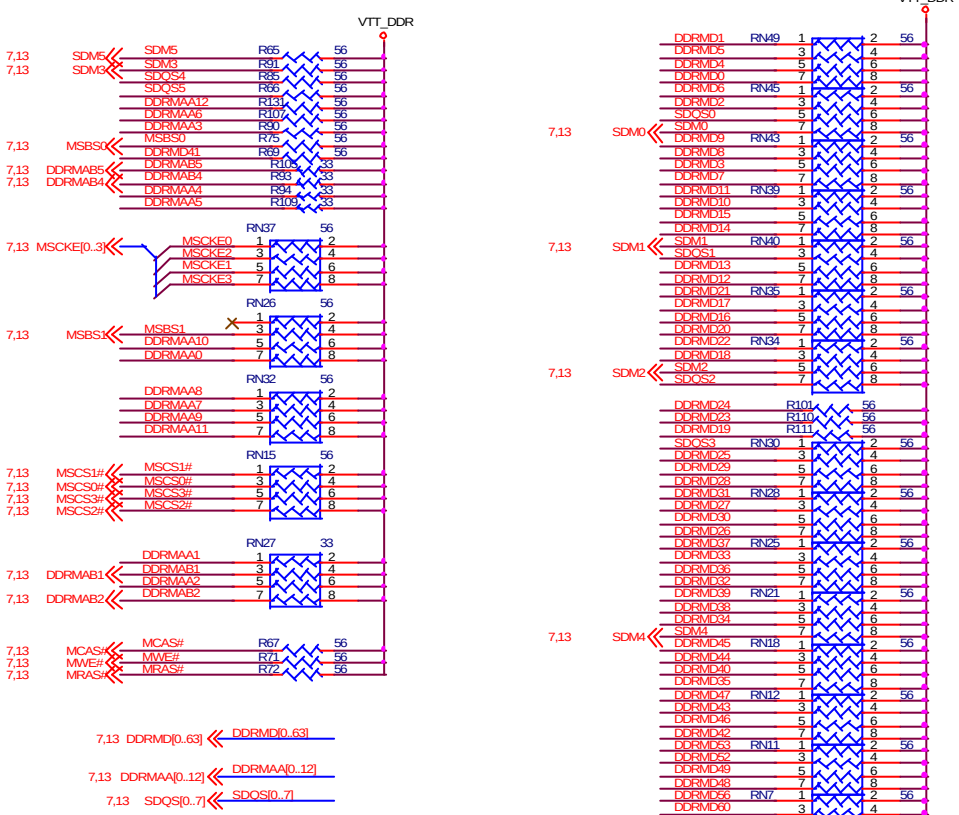
MICRO-STAR

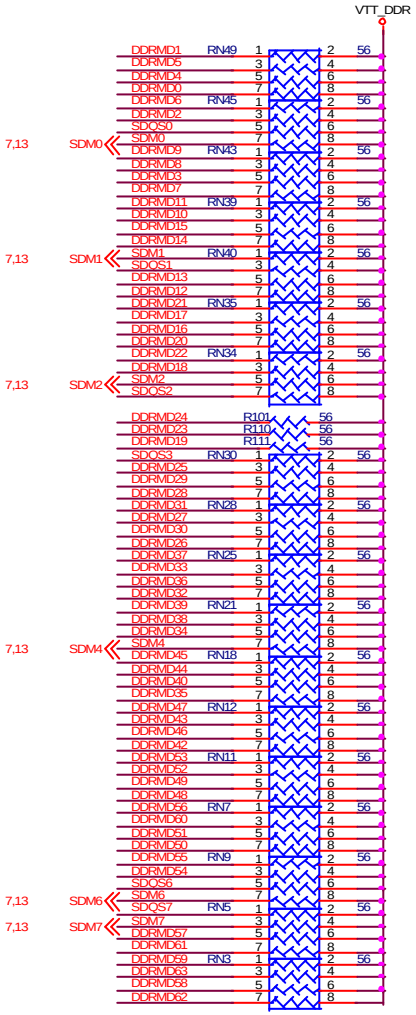
Title: DDR DIMM1 & 2

Size | Document Number: MS-7057 | Rev: 100

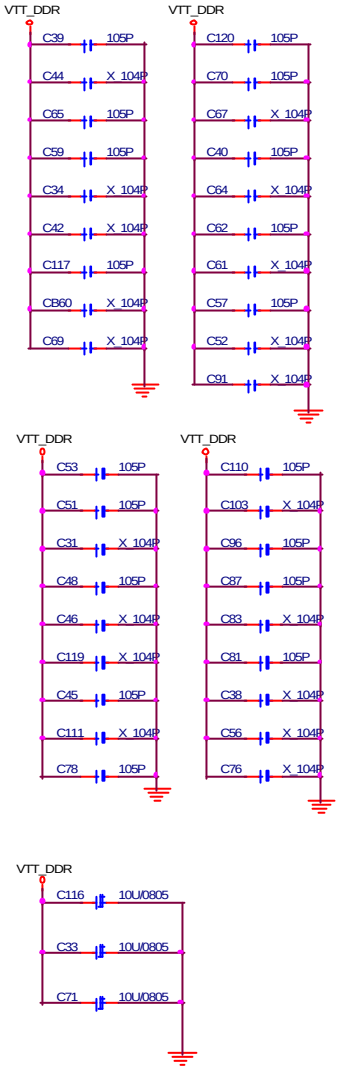
Date: Monday, February 16, 2004 | Sheet 13 of 27

DDR TERMINATORS





TERMINATION DECOUPLING



MICRO-STAR

Title: DDR DAMPING

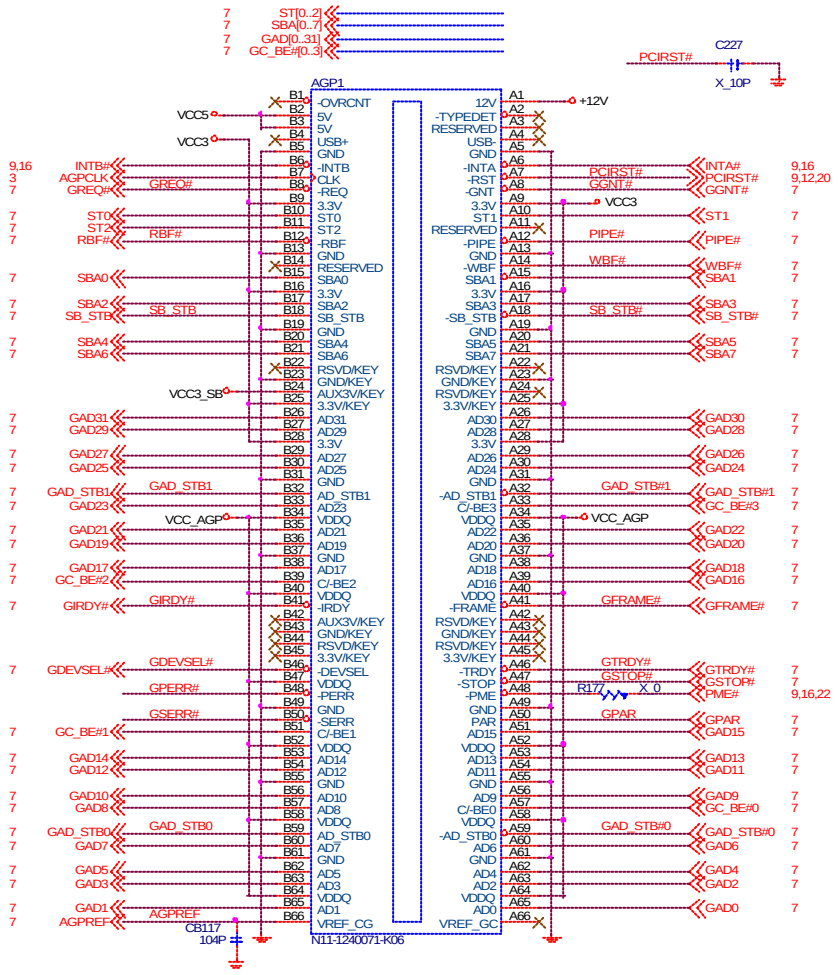
Size | Document Number: MS-7057 | Rev: 100

Date: Monday, February 16, 2004 | Sheet 14 of 27

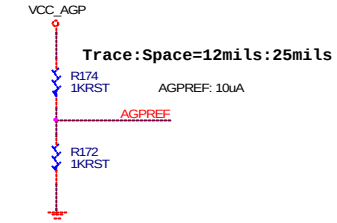
Changed Footprint : AGP1_5V_29mil

AGP UNIVERSAL 2X/4X SLOT(AGP VER:2.0 COMPLY)

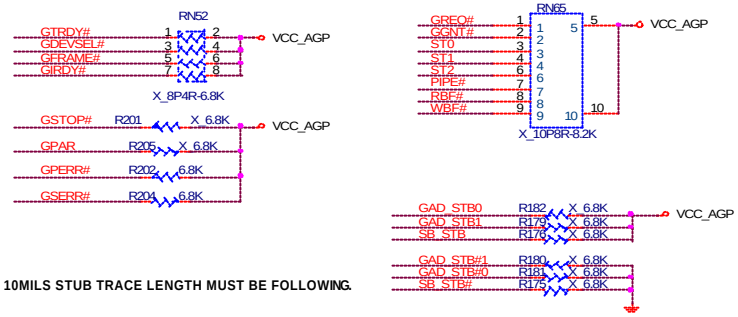
VCC5 = 60mils trace / 15 mils space



AGP SIGNAL REFERENCE CIRCUIT

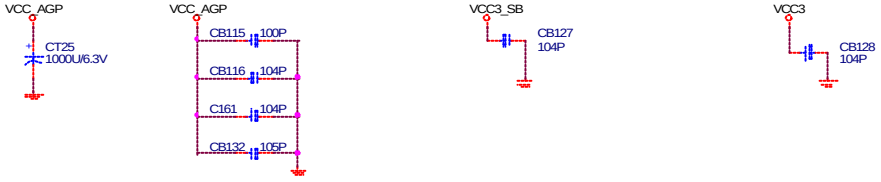


AGP TERMINATION RESISTORS



LESS 10MILS STUB TRACE LENGTH MUST BE FOLLOWING.

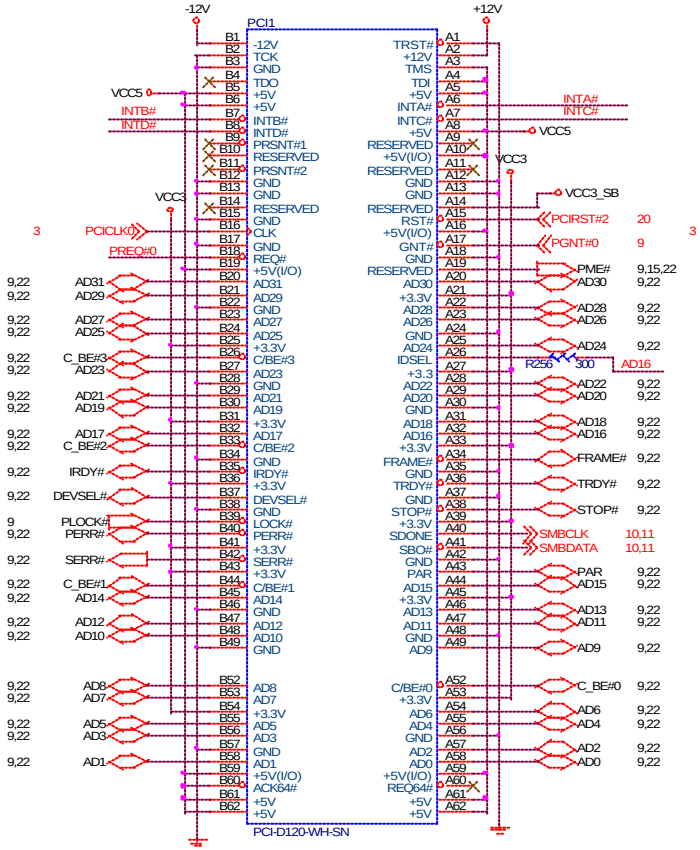
AGP SLOT DECOUPLING CAPACITORS



MICRO-STAR

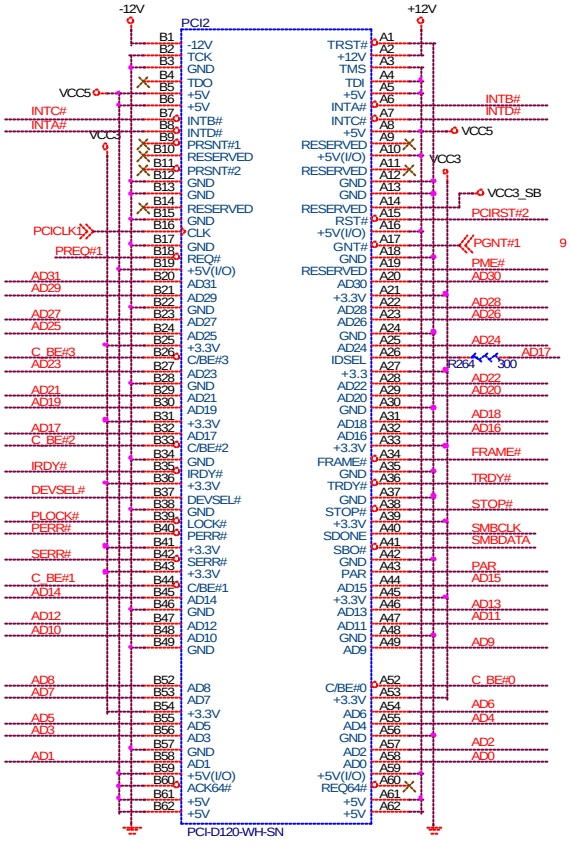
Title	AGP SLOT	
Size	Document Number MS-7057	Rev 100
Date: Monday, February 16, 2004	Sheet 15 of 27	

PCI SLOT 1 (PCI VER: 2.2 COMPLY)



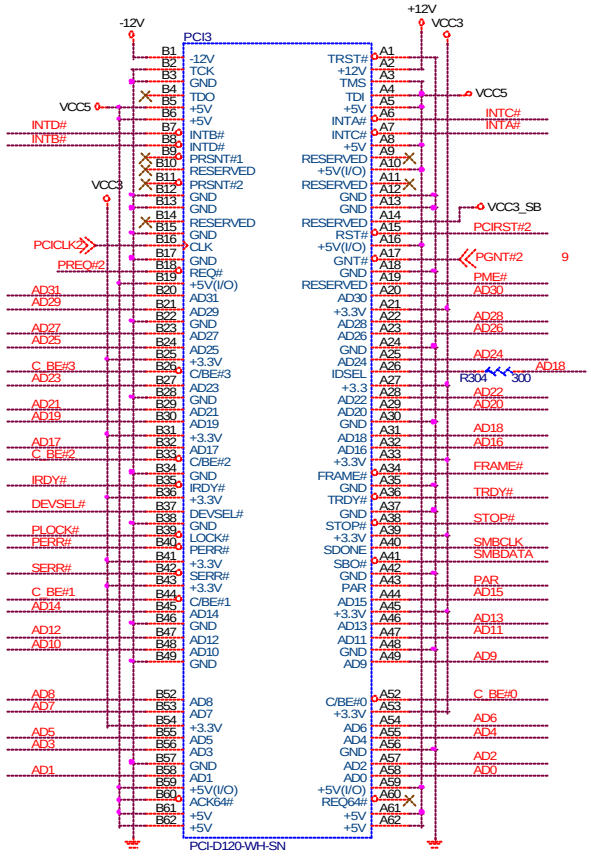
IDSEL = AD16
MASTER = PREQ0
INTA#

PCI SLOT 2 (PCI VER: 2.2 COMPLY)



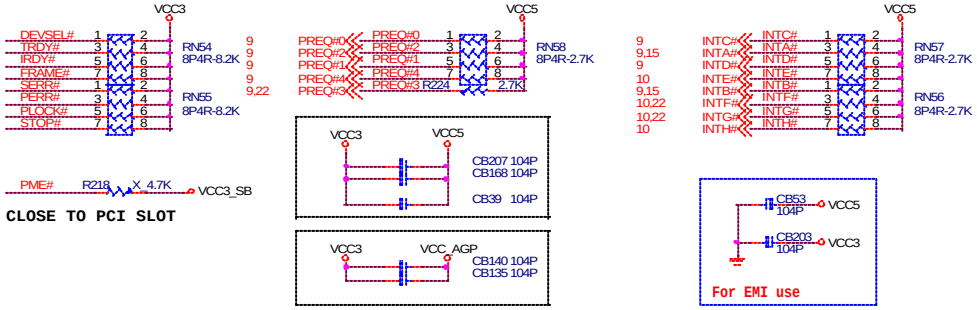
IDSEL = AD17
MASTER = PREQ1
INTB#

PCI SLOT 3 (PCI VER: 2.2 COMPLY)

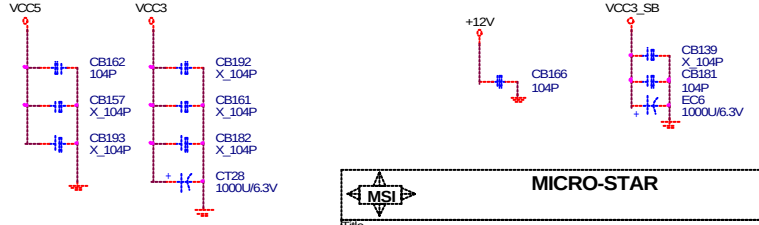


IDSEL = AD18
MASTER = PREQ2
INTC#

PCI PULL-UP / DOWN RESISTORS



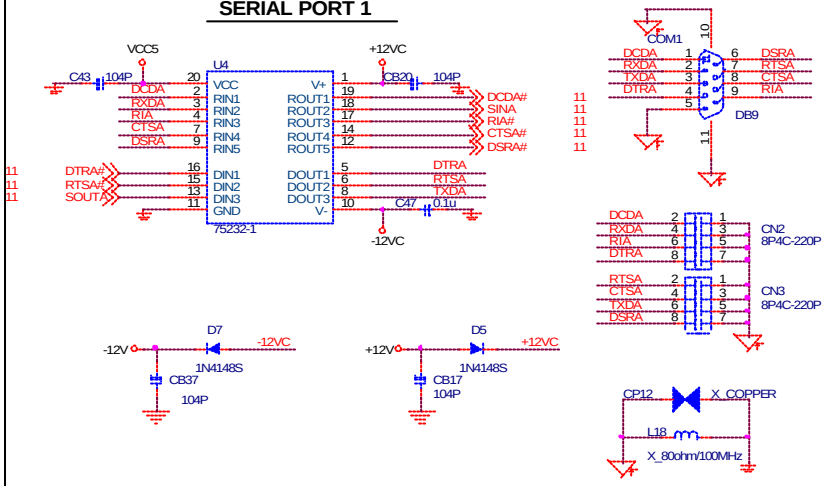
PCI SLOT DECOUPLING CAPACITORS



MSI
MICRO-STAR

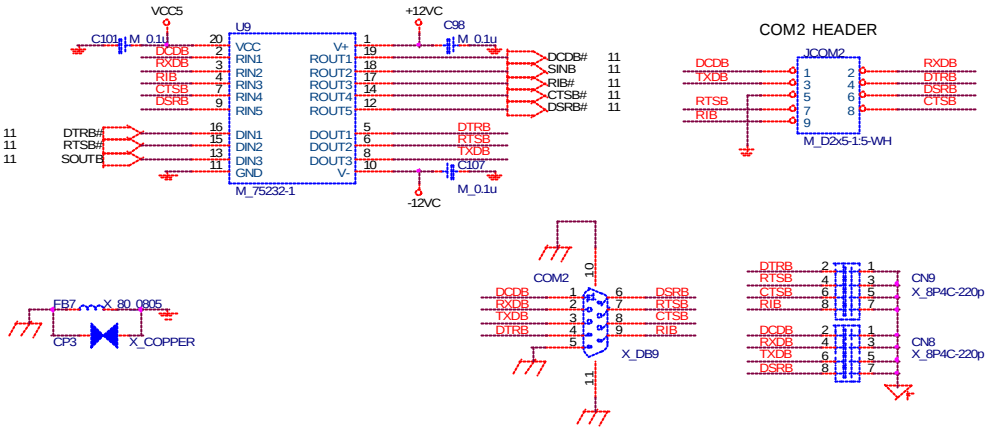
Title	PCI SLOT 1,2,3	
Size	Document Number: MS-7057	Rev: 100
Date: Monday, February 16, 2004	Sheet 16 of 27	

SERIAL PORT 1

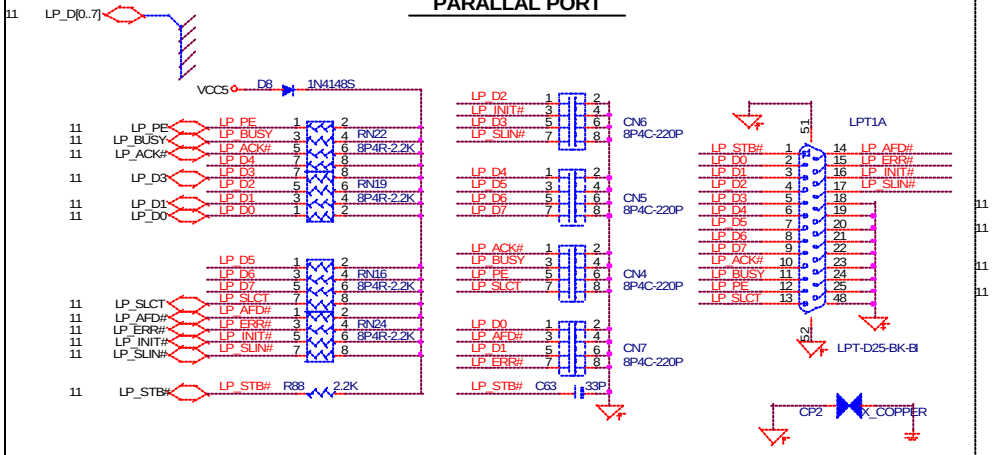


SERIAL PORT 2

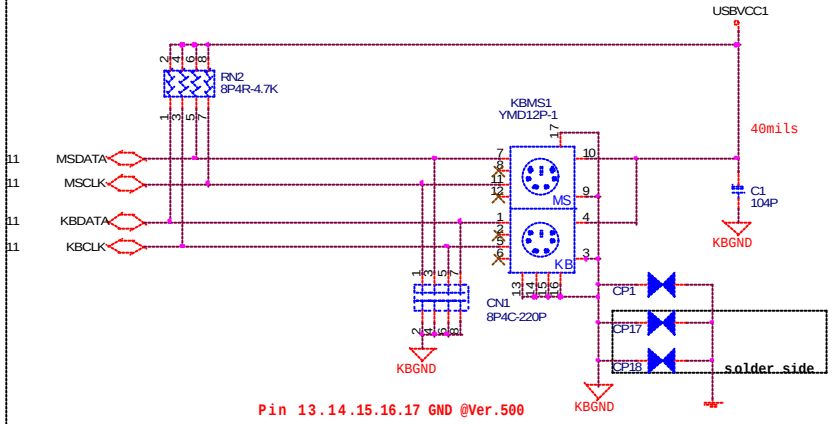
Take off CN11 & CN12



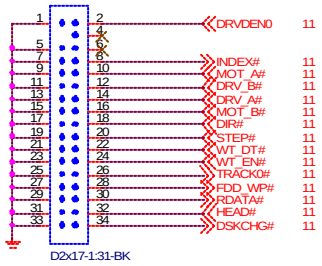
PARALLAL PORT

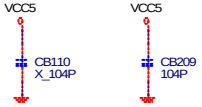


PS2 KEYBOARD & MOUSE CONNECTOR



FLOPPY CONNECTOR



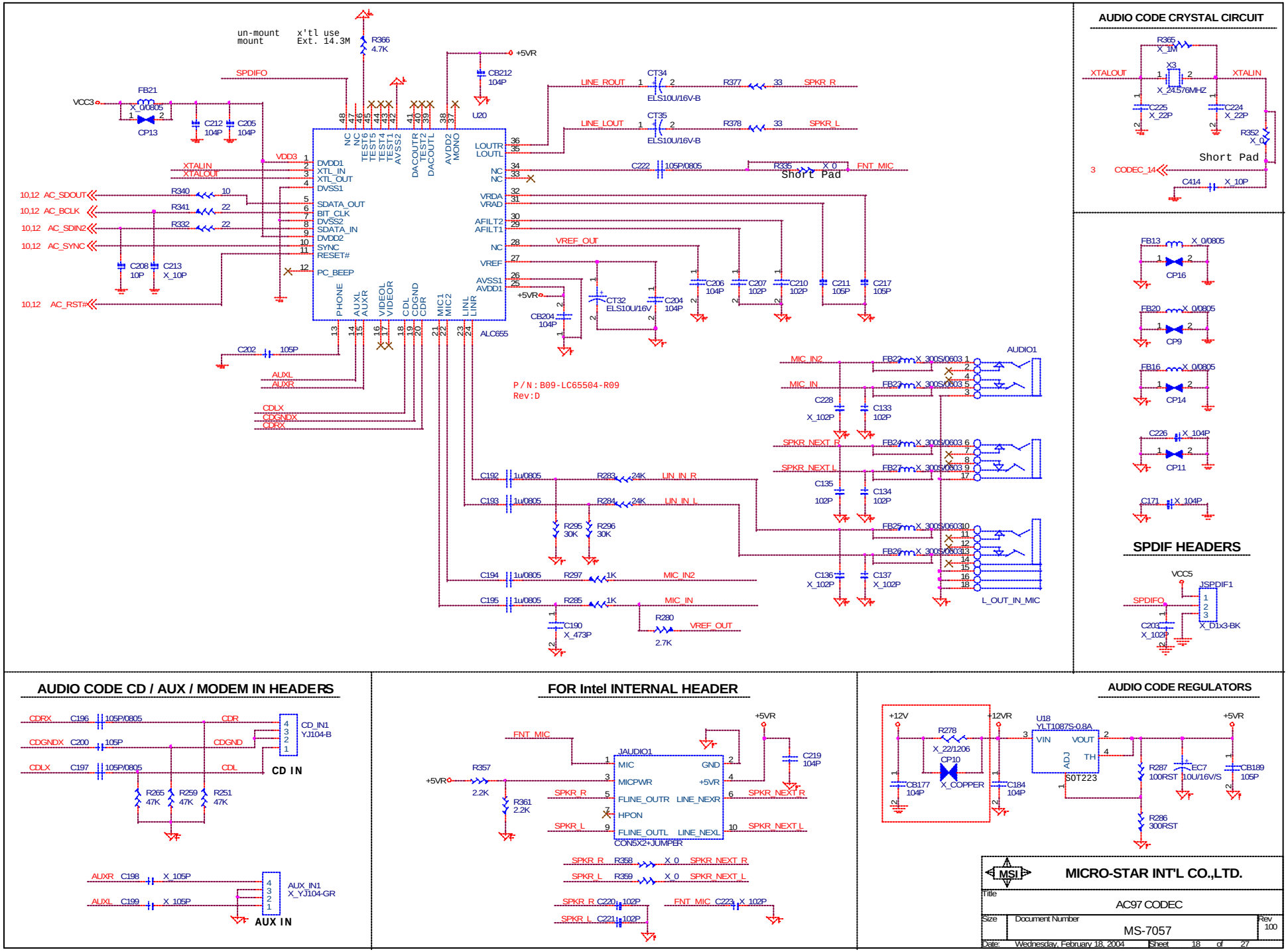


MSI MICRO-STAR

Title: I/O CONNECTORS

Size	Document Number	Rev
	MS-7057	100

Date: Tuesday, February 17, 2004 Sheet 17 of 27



AUDIO CODE CRYSTAL CIRCUIT

SPDIF HEADERS

AUDIO CODE CD / AUX / MODEM IN HEADERS

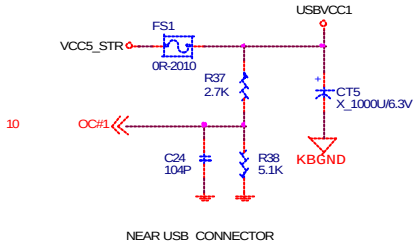
FOR Intel INTERNAL HEADER

AUDIO CODE REGULATORS

MSI	MICRO-STAR INT'L CO.,LTD.	
Title	AC97 CODEC	
Size	Document Number: MS-7057	Rev: 100
Date: Wednesday, February 18, 2004	Sheet 18 of 27	

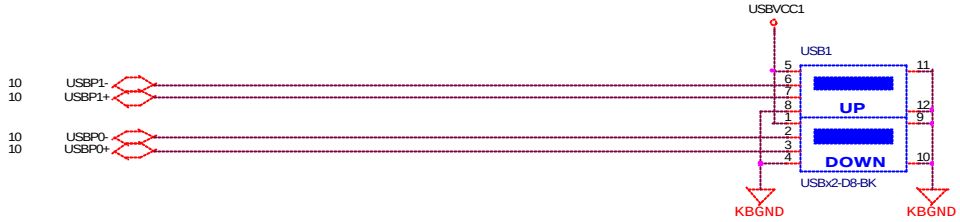
POWER CIRCUIT FOR USB PORT 0,1

Take off C28 & C100



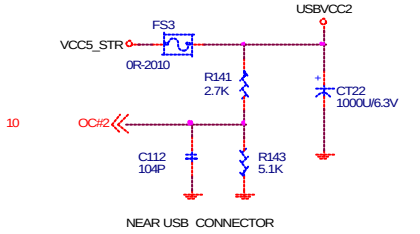
NEAR USB CONNECTOR

REAR PANEL USB CONNECTOR FOR USB PORT 0,1



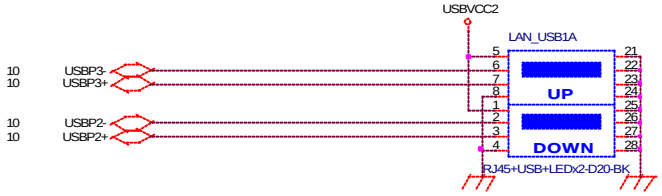
NEAR USB CONNECTOR

POWER CIRCUIT FOR USB PORT 2,3



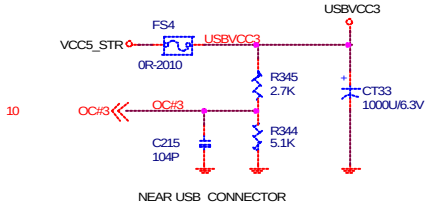
NEAR USB CONNECTOR

REAR PANEL USB CONNECTOR FOR USB PORT 2,3



NEAR USB CONNECTOR

POWER CIRCUIT FOR USB PORT 4,5

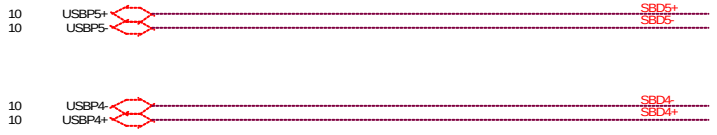


NEAR USB CONNECTOR

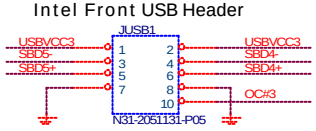
*** USB Trace width : 7.5 mils**
*** USB Trace Spacing : 20 mils**

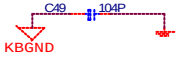
*** Differential USB Signlas Trace, Spacing : 7.5 mils**
*** USB Power Trace must be 50mils width**

FRONT PANEL USB CONNECTOR FOR USB PORT 4,5



Intel Front USB Header

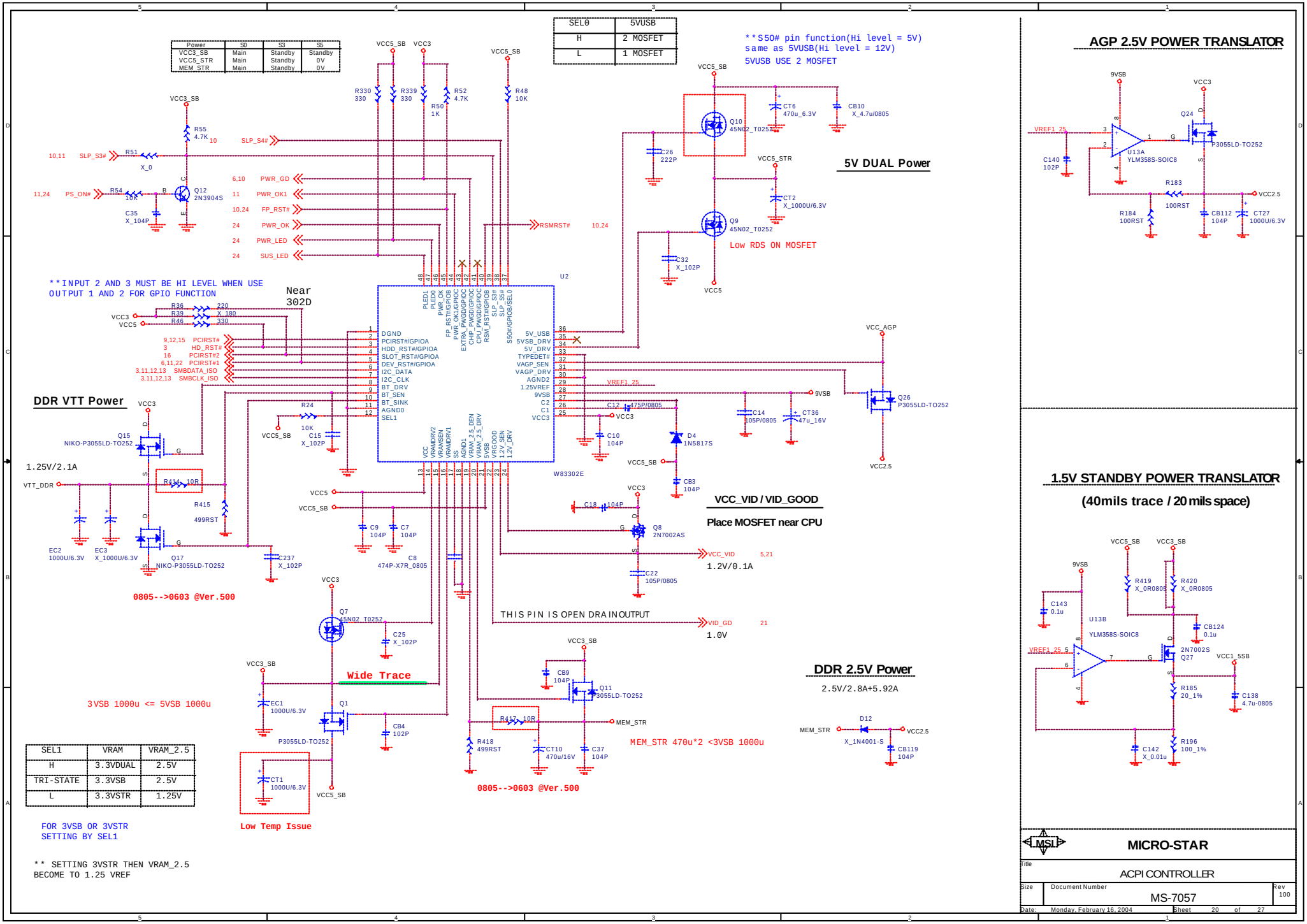


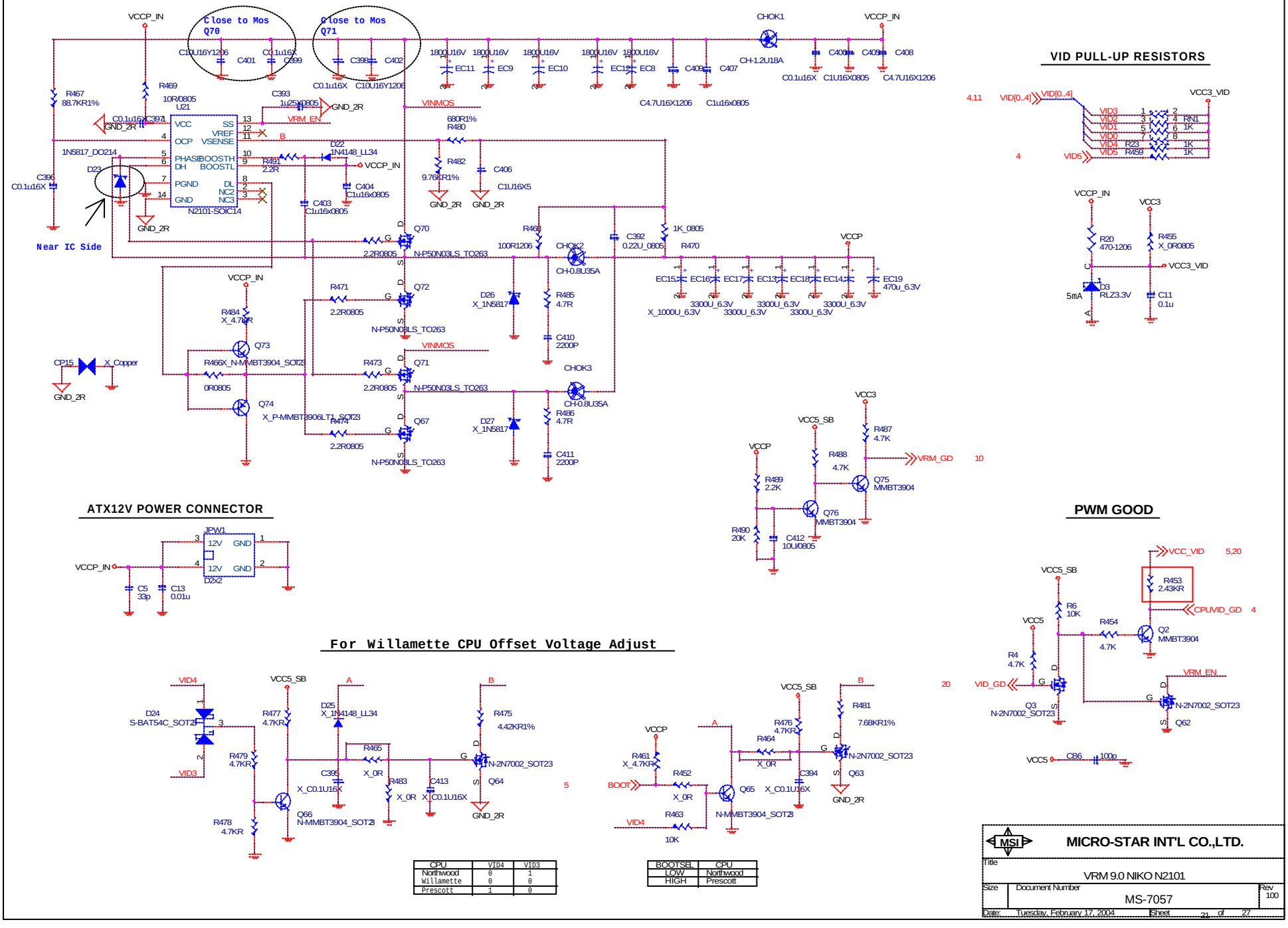


NEAR USB CONNECTOR

MSI **MICRO-STAR**

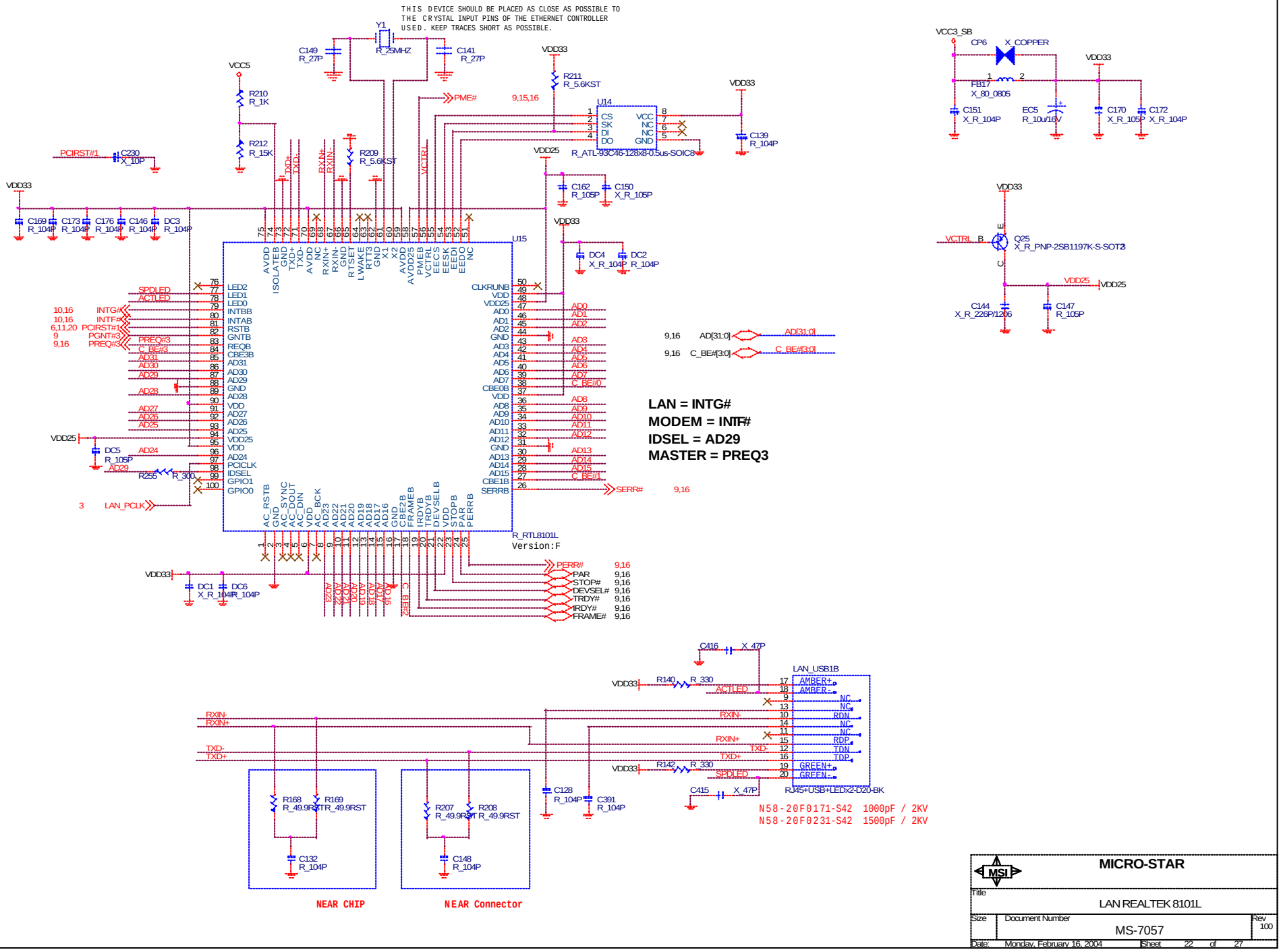
Title	USB CONNECTORS	
Size	Document Number MS-7057	Rev 100
Date: Monday, February 16, 2004	Sheet 19 of 27	


Power | S0 | S3 | S5
VCC3_SB | Main | Standby | Standby
VCC5_STR | Main | Standby | 0V
MEM_STR | Main | Standby | 0V
SEL0 | 5VUSB
H | 2 MOSFET
L | 1 MOSFET
**S5O# pin function(Hi level = 5V)
same as 5VUSB(Hi level = 12V)
5VUSB USE 2 MOSFET
5V DUAL Power
Low RDS ON MOSFET
**INPUT 2 AND 3 MUST BE HI LEVEL WHEN USE
OUTPUT 1 AND 2 FOR GPIO FUNCTION
Near 302D
DDR VTT Power
1.25V/2.1A
0805-->0603 @Ver.500
VCC_VID / VID_GOOD
Place MOSFET near CPU
1.2V/0.1A
THIS PIN IS OPEN DRAIN OUTPUT
1.0V
Wide Trace
3VSB 1000u <= 5VSB 1000u
MEM_STR 470u*2 <3VSB 1000u
DDR 2.5V Power
2.5V/2.8A+5.92A
SEL1 | VRAM | VRAM_2.5
H | 3.3VDUAL | 2.5V
TRI-STATE | 3.3VSB | 2.5V
L | 3.3VSTR | 1.25V
FOR 3VSB OR 3VSTR
SETTING BY SEL1
** SETTING 3VSTR THEN VRAM_2.5
BECOME TO 1.25 VREF
Low Temp Issue
AGP 2.5V POWER TRANSLATOR
1.5V STANDBY POWER TRANSLATOR
(40mils trace / 20mils space)
MICRO-STAR
ACPI CONTROLLER
MS-7057
Monday, February 16, 2004
Sheet 20 of 27



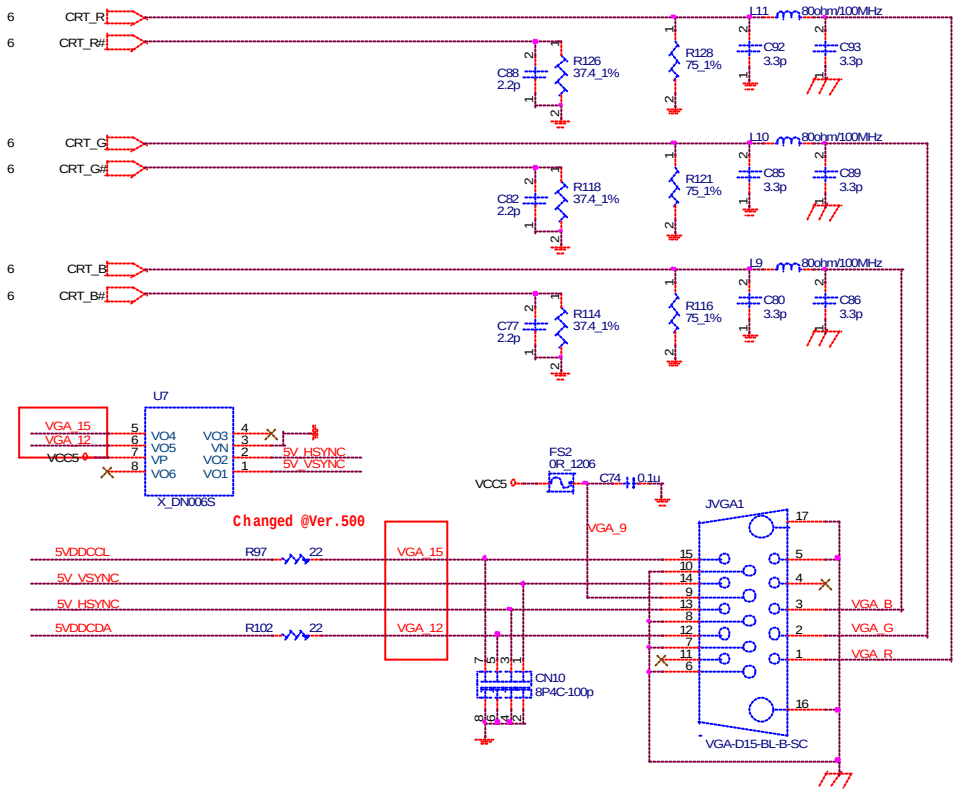
CPU	VID4	VID3
Northwood	0	1
Willamette	0	0
Prescott	1	0

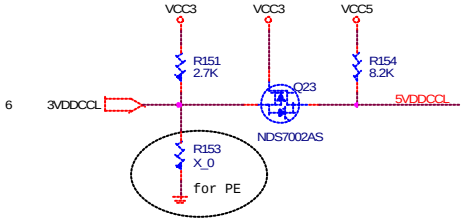
BOOTSEL	CPU
LOW	Northwood
HIGH	Prescott

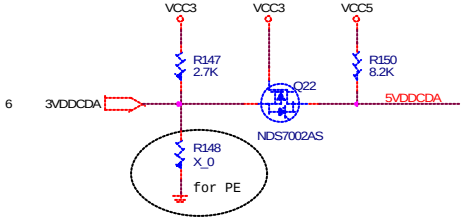


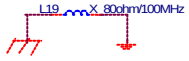
MSI	MICRO-STAR		
Title	LAN REALTEK 8101L		
Size	Document Number	MS-7057	Rev 100
Date: Monday, February 16, 2004	Sheet 22 of 27		

Video Connector

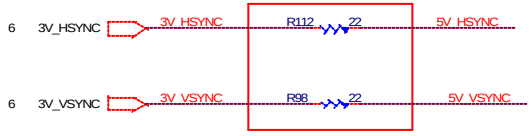








Took Off U8 & U6 @Ver.500



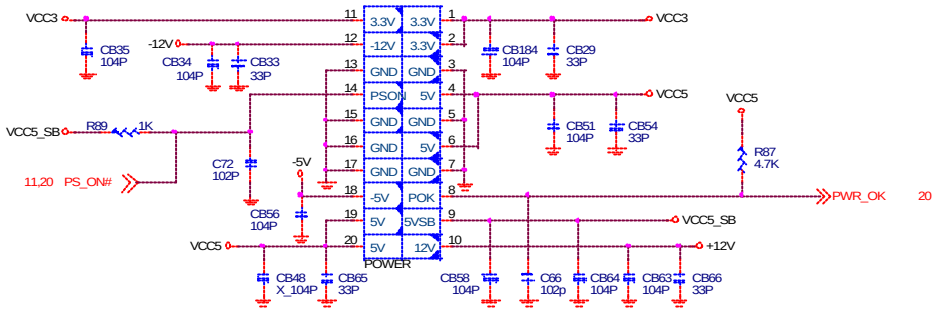
MSI **MICRO-STAR**

Title: VGA CONNECTOR

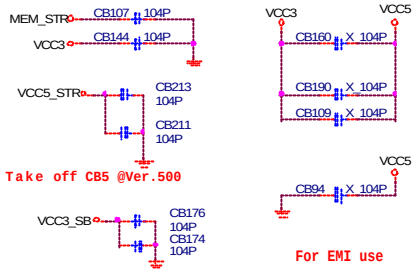
Size	Document Number	Rev
	MS-7057	100

Date: Tuesday, February 17, 2004 Sheet 23 of 27

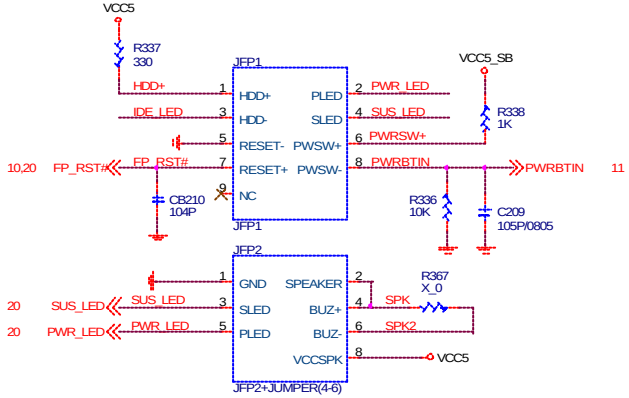
ATX CONNECTOR



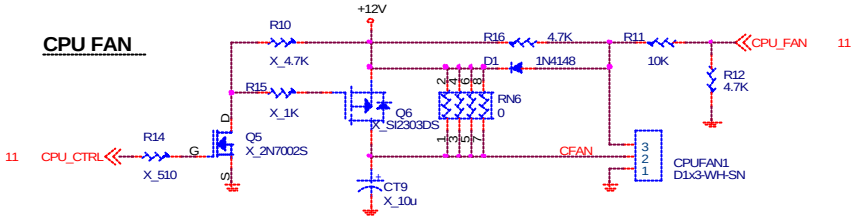
REGULATORS OUTPUT DECOUPLING CAPACITORS



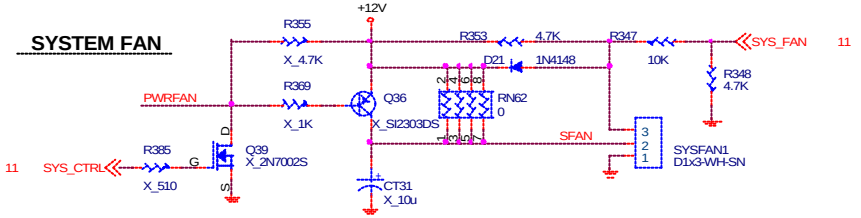
Intel Front Panel

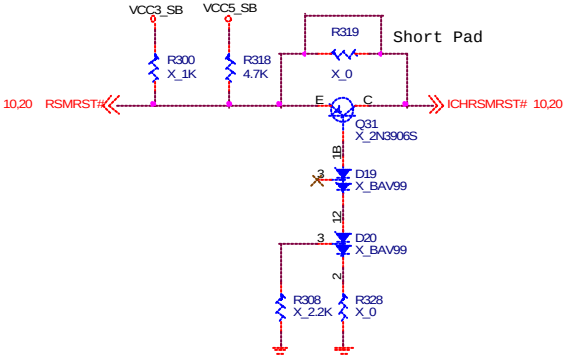


CPU FAN

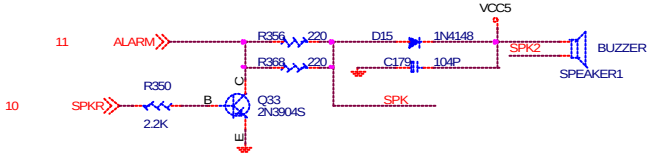


SYSTEM FAN

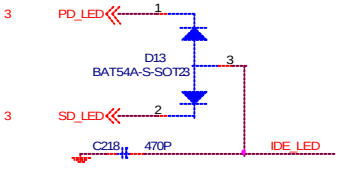




BUZZER



IDE LED



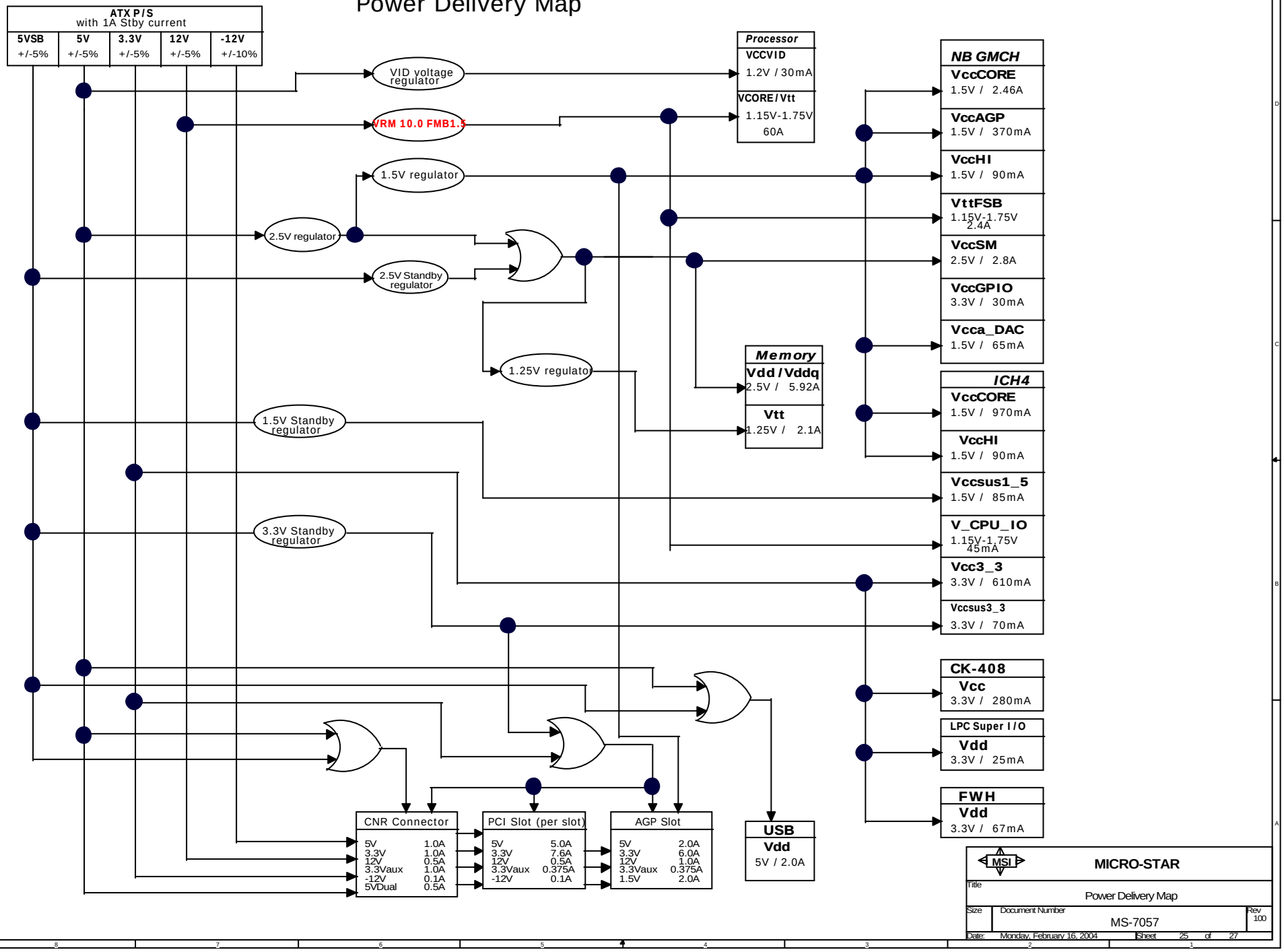
MSI **MICRO-STAR**

Title: Front Panel & ATX Connector & FAN

Size | Document Number: MS-7057 | Rev: 100

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Power Delivery Map



MSI MICRO-STAR

Title: Power Delivery Map

Size | Document Number: MS-7057 | Rev: 100

Date: Monday, February 16, 2004 | Sheet 25 of 27

General SPEC

ICH4

GPIO Pin	Type	Function
GPIO 0	I	REQ#A (multifunction pin)
GPIO 1	I	REQ#B (multifunction pin)
GPIO 2	I	Pull up through 8.2K ohms (PIRQE#)
GPIO 3	I	Pull up through 8.2K ohms (PIRQF#)
GPIO 4	I	Pull up through 8.2K ohms (PIRQG#)
GPIO 5	I	Pull up through 8.2K ohms (PIRQH#)
GPIO 6	I	Pull down through 10K ohms (unused)
GPIO 7	I	Pull down through 10K ohms (unused)
GPIO 8	I	Pull Up to 3.3VSBY through 4.7K ohms (SIO_PME)
GPIO 9	I	Not Implemented
GPIO 10	I	Not Implemented
GPIO 11	I	SMB_ALERT (multifuntion pin)
GPIO 12	I	EXTSMI# with Pull up 10K ohms to VCC3_SB
GPIO 13	I	Pull down through 10K ohms (unused)
GPIO 14~15	I	Not Implemented
GPIO 16	O	GNT#A (multifunction pin)
GPIO 17	O	GNT#B (multifuntion pin)
GPIO 18*	O	No Connected
GPIO 19	O	No Connected
GPIO 20	O	No Connected
GPIO 21	O	No Connected
GPIO 22	OD	No Connected
GPIO 23	O	Pull Up to 3.3V through 8.2K ohms (BIOS protect)
GPIO 24	I/O	GPIO 24
GPIO 25	I/O	GPIO 25
GPIO 26	I/O	Not Implemented
GPIO 27	I/O	No Connected
GPIO 28	I/O	No Connected
GPIO 29~31	O	Not Implemented
GPIO 32	I/O	No Connected
GPIO 33	I/O	No Connected
GPIO 34	I/O	Primary IDE ATA66/100 detection (PD_DET)
GPIO 35	I/O	Secondary IDE ATA66/100 detection (SD_DET)
GPIO 36	I/O	No Connected
GPIO 37	I/O	No Connected
GPIO 38	I/O	No Connected
GPIO 39	I/O	No Connected
GPIO 40	I/O	No Connected
GPIO 41	I/O	No Connected
GPIO 42	I/O	No Connected
GPIO 43	I/O	No Connected
GPIO 44~47	I/O	Not Implemented

** GPIO18 will toggle at 1Hz frequency.*

FWH

GPIO Pin	Type	Function
GPI 0	I	Pull down through 8.2K ohms (unused)
GPI 1	I	Pull down through 8.2K ohms (unused)
GPI 2	I	P1 customer defined
GPI 3	I	P1 customer defined
GPI 4	I	Pull down through 8.2K ohms (unused)

PCI Config.

DEVICE	ICH INT Pin	IDSEL	CLOCK	CLK GEN PIN OUT
PCI Slot 1	INTA# INTB# INTC# INTD#	AD16	PCICLK0	10 (PCI3/FS4)
PCI Slot 2	INTB# INTC# INTD# INTA#	AD17	PCICLK1	11 (PCI4)
PCI Slot 3	INTC# INTD# INTA# INTB#	AD18	PCICLK2	12 (PCI5)
PCI LAN	INTG# INTF#	AD29	LAN_PCLK	17 (PCI9)

**ICH4 reserved PCI address line AD22 for the PCI-to-ISA Bridge's IDSEL input.*

DIMM Config.

DEVICE	ADDRESS	CLOCK
DIMM 1	1010000B	DCLK0/DCLK 0# DCLK1/DCLK 1# DCLK2/DCLK 2#
DIMM 2	1010001B	DCLK3/DCLK 3# DCLK4/DCLK 4# DCLK5/DCLK 5#

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Jumper Setting & Connector Setting

PCB1
P01-7057100

BAT1_X
BATTERY

Q70_H1
MOS-HEATSINK

Q71_H1
MOS-HEATSINK

U10_1
E31-0400741-K08

U19_1
FLASH 4M

U5_1
CPU Retention

JMCH1
FAN Header

JMCH2
FAN Header

LABEL1
G51-LA00830

STD	U10_A B01-845GE15-I06 **845GE B1** U17_A B01-801DB55-I06 **ICH4 B0** LAN_USB1_A USB X 2 - 8pin Only USB*2	U10_A1 E31-0400741-K08 U19_A1 FLASH 4M LABEL1_A G51-LA00830 **845GE**
LAN	LAN_USB1_B USB X2 +RJ45 N58-20F0171-S42 1000pF / 2KV N58-20F0231-S42 1500pF / 2KV LAN & Tramsformer + USB*2	R135_A 137_1% R126_A 37.4_1% R118_A 37.4_1% R114_A 37.4_1% R128_A 75_1% R121_A 75_1% R116_A 75_1% Onboard VGA :Yes
GV	U10_B B01-845GV15-I06 **845GV B1** U17_B B01-801DB55-I06 **ICH4 B0**	U19_B1 FLASH 2M LABEL1_B G51-LA00828 **845GV**
G	U10_C B01-0845G65-I06 **845G B1** U17_C B01-801DB55-I06 **ICH4 B0**	LABEL1_C G51-LA00829 **845G(B)**
GL	U10_D B01-845GL15-I06 **845GL B1** U17_D B01-801DB55-I06 **ICH4 B0**	U10_B1 E31-0400440-A02 LABEL1_D G51-LA00831 **845GL**
B	U10_E B01-845PE15-I06 **845PE B1** U17_E B01-801DB55-I06 **ICH4 B0**	CB92_B 0 ohm R135_B 0 C160_B 0 R126_B 0 R118_B 0 R114_B 0 R128_B 0 R121_B 0 R116_B 0 Onboard VGA :No
OPT:A **OPT:B**		L9_A 600ohm/100MHz L10_A 600ohm/100MHz L11_A 600ohm/100MHz C80_A 10p C86_A 10p C85_A 10p C89_A 10p C92_A 10p C93_A 10p

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Title: JUMPER SETTING & MANUAL PARTS

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